

## STATISTICAL ANALYSIS PLAN (SAP)

### BNT324-01

|                                  |                                                                                                                                                                                                            |                             |             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|
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| Sponsor Name:                    | BioNTech SE                                                                                                                                                                                                |                             |             |
| Protocol Number:                 | BNT324-01                                                                                                                                                                                                  |                             |             |
| Protocol Title:                  | A Phase Ib/II, multi-site, open-label, two-part trial to evaluate the efficacy, safety, pharmacokinetics, and recommended combination dose of BNT324 with BNT327 in participants with advanced lung cancer |                             |             |
| Protocol Version:                | 1.0_EU                                                                                                                                                                                                     |                             |             |
| Protocol Version Date:           | 29 AUG 2025                                                                                                                                                                                                |                             |             |
| Compound Name:                   | BNT324+BNT327                                                                                                                                                                                              |                             |             |
| Study Phase:                     | Phase Ib/II                                                                                                                                                                                                |                             |             |
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## SAP VERSION HISTORY

| SAP Version | Approval Date | Based on Protocol (Version, Approval Date) | Brief Description of Change(s)/Revision(s) and Rationale |
|-------------|---------------|--------------------------------------------|----------------------------------------------------------|
| 1.0         | 05 MAR 2026   | Version 1.0_EU, 29 AUG 2025                | Original document creation.                              |

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## 1 INTRODUCTION

This is a Phase Ib/II, multi-site, open-label, two-part trial designed to evaluate and establish two safe combination dose levels (RP2D and RP2D-1) of BNT324 with BNT327 (Part 1), to determine the optimal combination dose (dose optimization [DO]) in NSCLC and SCLC lead indication cohorts at the RP2D and RP2D-1 and to evaluate the preliminary efficacy in selected lung cancer cohorts at the highest combination dose level (in a signal seeking Part 2). The results might be included in a regulatory submission. BNT324 can also be referred to as DB-1311.

This statistical analysis plan (SAP) describes the detailed procedures for the planned statistical analyses of the efficacy, safety, immunogenicity, pharmacokinetic, and patient-reported outcome (PRO) data that are collected. The endpoints are defined, and the statistical methods used to analyze them are presented. Further analysis of pharmacokinetic data and modeling will be detailed in a separate document. Table shells for the planned tables, figures, and listings (TFL) are included in a separate document, entitled "Data Presentation Plan".

This version of the SAP has been developed using the BNT324-01 clinical trial protocol (CTP) version 3.0, dated 04 FEB 2025. Changes to the methods planned in the CTP, if applicable, will be summarized in Section 7.

### 1.1 Trial Objectives, Endpoints and Estimands

Table 1: Objectives and endpoints

| OBJECTIVES                                                                                                                                     | ENDPOINTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary objective</b>                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Part 1 (Dose Escalation)</b>                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| To determine the RP2D of BNT324 in combination with BNT327 by assessing the safety and tolerability in participants with advanced lung cancer. | By dose level: <ul style="list-style-type: none"><li>• Occurrence of DLTs during the DLT evaluation period (Cycle 1, 21 days).</li><li>• Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.</li><li>• Occurrence of drug interruption, reduction, and treatment discontinuations due to TEAEs by dose level from the time of the first dose of IMP to 90 days after the last dose of IMP or until new anticancer therapy is started, whichever occurs first.</li></ul> |

|                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Part 2 (Dose Optimization/Signal Seeking)</b>                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><u>Lead indications Cohort 1 and Cohort 2:</u><br/>To determine the optimal dose of BNT324 in combination with BNT327 by assessing the safety profile and efficacy of the combination therapy in the randomized dose optimization cohorts in the lead cohorts (Cohort 1 treatment naive non-sq NSCLC and Cohort 2 relapsed/progressive SCLC).</p> | <p>By cohort and treatment arm:</p> <ul style="list-style-type: none"> <li>• Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.</li> <li>• Occurrence of drug interruption, reduction, and treatment discontinuation due to TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.</li> <li>• ORR, defined as the proportion of participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment).</li> </ul>                                                                                                                                                                                    |
| <p><u>Efficacy signal seeking Cohorts 3-7:</u><br/>To evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1.</p>                                                                                                                                                                                                        | <p>By cohort:</p> <ul style="list-style-type: none"> <li>• ORR, defined as the proportion of participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Secondary objectives</b>                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Part 1 (Dose Escalation)</b>                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>To evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1.</p>                                                                                                                                                                                                                                                        | <p>By dose level:</p> <ul style="list-style-type: none"> <li>• ORR, defined as the proportion of participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment).</li> <li>• DCR, defined as the proportion of participants with confirmed CR, PR, or SD as best overall response (per RECIST 1.1 based on the investigator's assessment).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Part 2 (Dose Optimization/Signal Seeking)</b>                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>To evaluate the efficacy (other than ORR) of BNT324 in combination with BNT327 in dose optimization/signal seeking cohorts 1-7.</p>                                                                                                                                                                                                               | <p>By cohort and treatment arm:</p> <ul style="list-style-type: none"> <li>• PFS defined as the time from first dose of IMP to the first objective tumor progression (PD) or death from any cause, whichever occurs first, per RECIST 1.1 based on the investigator's assessment.</li> <li>• DOR, defined as the time from first objective response (CR or PR) to first occurrence of objective tumor progression (PD) or death from any cause, whichever occurs first (per RECIST 1.1 based on the investigator's assessment).</li> <li>• OS, defined as the time from first dose of IMP to death from any cause.</li> <li>• DCR, defined as the proportion of participants with confirmed CR, PR, or SD as best overall response (per RECIST 1.1 based on the investigator's assessment).</li> <li>• TTR defined as the time from first dose of IMP to first objective response (CR or PR per RECIST 1.1 based on the investigator's assessment).</li> </ul> |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To assess the safety of BNT324 in combination with BNT327 in the signal seeking Cohorts 3-7. | <p>By cohort and treatment arm:</p> <ul style="list-style-type: none"> <li>• Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMPs to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.</li> <li>• Occurrence of drug interruption, reduction, and treatment discontinuation due to TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.</li> </ul>                                                                                                                                                            |
| <b>Exploratory objectives</b>                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>All parts and all cohorts</b>                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| To assess the PK of BNT324 (ADC, total antibody, and free payload P1021) and BNT327.         | <ul style="list-style-type: none"> <li>• Concentrations of BNT324 ADC, total antibody, free payload P1021, and BNT327 in the serum or plasma.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| To characterize the immunogenicity of BNT324 in combination with BNT327.                     | <ul style="list-style-type: none"> <li>• The prevalence and incidence of ADAs against BNT324.</li> <li>• The prevalence and incidence of ADAs against BNT327.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| To evaluate B7-H3 and PD-L1 expression in tumor tissue and its relationship with efficacy.   | <p>By cohort and overall:</p> <ul style="list-style-type: none"> <li>• Expression level of B7-H3 in tumor tissue and its correlation with efficacy.</li> <li>• Expression of PD-L1 in tumor tissue and its correlation with efficacy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>For Part 1 (Dose Escalation)</b>                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| To evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1.       | <ul style="list-style-type: none"> <li>• PFS defined as the time from first dose of IMP to the first objective tumor progression (PD per RECIST 1.1) or death from any cause, whichever occurs first (based on the investigator's assessment).</li> <li>• DOR, defined as the time from first objective response (CR or PR) to first occurrence of objective tumor progression (PD) or death from any cause, whichever occurs first (per RECIST 1.1 based on the investigator's assessment).</li> <li>• OS, defined as the time from first dose of IMP to death from any cause.</li> <li>• TTR defined as the time from first dose of IMP to first objective response (CR or PR per RECIST 1.1 based on the investigator's assessment).</li> </ul> |
| <b>For Part 2 - Cohort 1 and 2 only</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| To evaluate HRQoL endpoints using the EORTC QLQ-C30, EORTC QLQ-LC29, PGIS, and PGIC.         | <ul style="list-style-type: none"> <li>• Change from baseline in EORTC QLQ-C30 Global Health status/QoL scale, functional scales, and symptom scales/item.</li> <li>• Change from baseline in EORTC QLQ-LC29 symptom scales/item.</li> <li>• Change from baseline in PGIS overall status.</li> <li>• Change from baseline in PGIS overall status.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                       |



## 1.2 Study Design

This is a Phase Ib/II, multi-site, open-label, two-part trial designed to evaluate and establish two safe combination dose levels (RP2D and RP2D-1) of BNT324 with BNT327 (Part 1), to determine the optimal combination dose (dose optimization [DO]) in NSCLC and SCLC lead indication cohorts at the RP2D and RP2D-1 and to evaluate the preliminary efficacy in selected lung cancer cohorts at the highest combination dose level (in a signal seeking Part 2).

Patients with recurrent, progressive as well as advanced, metastatic, treatment-naïve NSCLC and SCLC are eligible to participate in the trial.

The trial consists of a screening period, a treatment period, a safety follow-up period, and a long-term survival follow-up period.

The trial consists of two parts as outlined in [Figure 1](#).

### **Part 1: Dose escalation of BNT324 in combination with BNT327**

In Part 1 participants with histologically or cytologically confirmed relapsed or progressive lung cancer (both SCLC and NSCLC are eligible) will receive BNT324 in combination with BNT327 using a Bayesian optimal interval (BOIN) dose escalation design. The maximum sample size is nine for each dose level and 54 for the overall BOIN design.

The BOIN design for the combination of BNT324 and BNT327 will be as follows:

Dose levels for BNT324:

- DL0 = 4.5 mg/kg
- DL1 = 6 mg/kg
- DL2 = 9 mg/kg

Dose levels for BNT327:

- DL0 = 20 mg/kg
- DL1 = 30 mg/kg

The trial will start at Dose Level 0-0 (DL0-0) as a combination of BNT324 at 4.5 mg/kg Q3W and BNT327 at 20 mg/kg Q3W in cohorts of three participants each following a BOIN design (see details in the [Appendix 2](#)).

The escalation and de-escalation decisions will be guided by prespecified BOIN boundaries, which are determined based on the target toxicity rate of 0.27. The maximum sample size is nine for each dose level and 54 for the overall BOIN design. This strategy aims to identify the RP2D for the combination of BNT324 and BNT327. The trial will continue enrolling cohorts and making escalation/de-escalation decisions based on the observed dose-limiting toxicity (DLT) rates and the BOIN boundaries. Per sponsor's decision, among the dose levels that have been demonstrated to be safe and well tolerated, or at their lower dose levels, additional up to 20 participants per dose level for each line (first line and second line) of NSCLC disease indication and second line SCLC disease indication may be backfilled, if they meet the criteria for Part 2. In total, up to 80 participants in each line (first line and second line) of NSCLC disease indication and up to 60 participants in the second line SCLC disease indication may be backfilled. Backfill participants will not be counted towards the maximum sample size of the BOIN design. An IRC will regularly review the trial data to ensure participant safety and to make recommendations regarding trial continuation, modification, or termination. Per IRC

decision, depending on cumulative safety data, alternative doses of BNT324 or BNT327 may be explored with up to nine DLT-evaluable participants for each dose level.

The decision to conduct a dose escalation for both BNT324 and BNT327 takes into consideration that a combination of novel agents may potentially result in additive toxicities or PK interactions. Therefore, the trial aims to identify the doses that demonstrate the optimal balance between the safety profile and potential efficacy.

The RP2D of the combination will be based on considerations of the overall assessment of safety data and available efficacy, PK, PD, ADA, and additional biomarker data, as applicable.

## **Part 2: Randomized dose optimization (DO) cohorts and signal seeking cohorts**

In Part 2 of the trial, BNT324 will be studied in combination with BNT327 at the RP2D compared to a lower/another combination dose level (RP2D-1) in participants with advanced metastatic treatment-naïve NSCLC (DO Cohort 1) and relapsed/progressive SCLC after failure of cytotoxic chemotherapy with or without IO (DO Cohort 2).

Approximately 40 participants will be randomly assigned at a 1:1 ratio to one of the two dose optimization levels (RP2D and RP2D-1) in each DO Cohort. The totality of the available data (e.g., safety, efficacy, PK etc.) will be reviewed to select the optimal dose. After the optimal dose is selected, an additional 20 participants in each cohort may be enrolled in the selected optimal dose.

In the signal seeking cohorts (Cohort 3-7), up to 40 participants in each cohort will receive BNT324 in combination with BNT327 at the RP2D or RP2D-1 from Part 1.

For each of the Cohorts 3-7, an early futility check on ORR results will be performed after the first 10 participants become response evaluable (see Section 3 for the definition of the Response Evaluable Set) in the corresponding cohort. The cohort may be considered futile and stopped for further enrollment if it doesn't meet the criteria as listed in Table 3. Enrollment will not pause when conducting the futility check.

After discontinuation of trial treatment, all participants will have a post-treatment safety follow-up scheduled at 90 days after their last dose of IMP ( $\pm 10$  days) or up to 10 days prior to start of subsequent therapy, whichever occurs first. Participants who have discontinued treatment for reasons other than PD will also be followed up with tumor assessments until radiological progression (or death). Participants who continue treatment beyond progression should continue undergoing tumor scans according to the SoA timelines, despite having recorded PD. All participants should be followed up for survival unless consent is withdrawn. Long-term/Survival Follow-up visits will be performed every 3 months ( $\pm 14$  days) from the Safety Follow-up visit until death, withdrawal of consent, or trial closure, whichever occurs first.

A summary of cohorts, IMPs, dose regimens, and sample size for Part 1 and Part 2 is provided in Figure 1 and Table 2, respectively.

**Table 2: Overview of cohorts, indications, and trial treatments in Part 2**

| Cohort <sup>a</sup> | Arm                 | Indication                                                            | IMP: Dose <sup>b</sup>        | No. of trial participants |
|---------------------|---------------------|-----------------------------------------------------------------------|-------------------------------|---------------------------|
| 1<br>DO             | RP2D <sup>a</sup>   | Non-sq NSCLC 1L, AGA neg.<br>any PD-L1 expression,<br>treatment-naïve | RP2D of BNT324 + BNT327 Q3W   | 20 to 40                  |
|                     | RP2D-1 <sup>a</sup> |                                                                       | RP2D-1 of BNT324 + BNT327 Q3W | 20 to 40                  |
| 2<br>DO             | RP2D <sup>c</sup>   | SCLC, 2L, post-chemo<br>+/- IO                                        | RP2D of BNT324 + BNT327 Q3W   | 20 to 40                  |
|                     | RP2D-1 <sup>c</sup> |                                                                       | RP2D-1 of BNT324 + BNT327 Q3W | 20 to 40                  |
| 3                   | RP2D                | Non-sq NSCLC AGA neg., 2L, post-chemo +/- IO                          | RP2D of BNT324 + BNT327 Q3W   | 40                        |
| 4                   | RP2D                | Sq NSCLC AGA neg., treatment-naïve                                    | RP2D of BNT324 + BNT327 Q3W   | 40                        |
| 5                   | RP2D                | Sq NSCLC AGA neg., 2L, post-chemo +/- IO                              | RP2D of BNT324 + BNT327 Q3W   | 40                        |
| 6                   | RP2D                | Non-sq NSCLC, AGA+, post-TKI                                          | RP2D of BNT324 + BNT327 Q3W   | 40                        |
| 7                   | RP2D                | ES-SCLC, treatment-naïve <sup>d</sup>                                 | RP2D of BNT324 + BNT327 Q3W   | 40                        |

- a) The sponsor may decide to not activate and/or close one or more of the cohorts.
- b) Depending on cumulative safety data, alternative doses of BNT324 and BNT327 may be explored.
- c) The doses for the two treatment arms for DO in Cohort 1 and Cohort 2 of Part 2 will be confirmed based on the data from Part 1. Up to 40 participants will be enrolled in each cohort and randomized at 1:1 ratio to one of the two different dose levels. After the optimal dose is selected, an additional 20 participants in each cohort may be enrolled in the selected optimal dose.
- d) Optional Cohort 7 may be initiated upon the EMA deeming the submitted data acceptable.

Abbreviations: AGA = actionable oncogenic alteration; chemo = chemotherapy; DO = dose optimization; ES = extended stage; IMP = investigational medicinal product; IO = immuno-oncology; L = line of therapy; neg = negative; (non-) sq = (non-) squamous; NSCLC = non-small cell lung cancer; PD-L1 = programmed death-ligand 1; Q3W = every 3 weeks; RP2D = recommended Phase II dose; SCLC = small cell lung cancer; Sq = squamous; TKI = tyrosine kinase inhibitor.

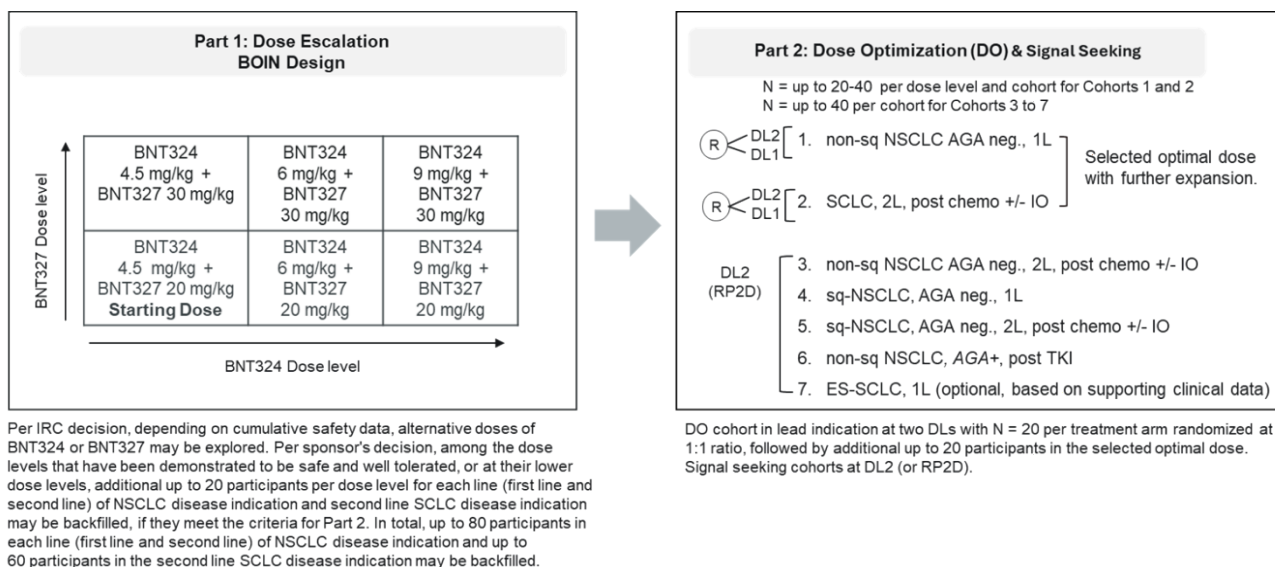
## Number of trial participants

It is anticipated that the trial will enroll up to 594 participants, including backfill participants in Part 1.

In Part 1, up to 54 DLT-evaluable participants will be enrolled following the BOIN design. Per sponsor's decision, among the dose levels that have been demonstrated to be safe and well tolerated, or at their lower dose levels, additional up to 20 participants per dose level for each line (first line and second line) of NSCLC disease indication and second line SCLC disease indication may be backfilled, if they meet the criteria for Part 2. In total, up to 80 participants in each line (first line and second line) of NSCLC disease indication and up to 60 participants in the second line SCLC disease indication may be backfilled. The backfilling participants will not be counted towards the total sample size of the BOIN portion of Part 1. A total of up to 274 participants will be enrolled in Part 1, including backfill participants.

In Part 2, up to ~40 participants in each cohort will be randomized at a 1:1 ratio to one of the two dose levels in the DO Cohort 1 and DO Cohort 2. The totality of the available data (e.g., safety, efficacy, PK etc.) will be reviewed to select the optimal dose. After the optimal

dose is selected, an additional 20 participants in each cohort may be enrolled in the selected optimal dose. Up to ~40 participants will be enrolled in each of the signal seeking disease Cohorts 3-7, for a total of up to 320 participants.



**Figure 1: Trial Schema**

[Appendix 1](#) displays the schedule of assessments for the study.

### 1.2.1 Treatment Randomization and/or Assignment and Blinding

This is an open-label trial, and blinding does not apply. Participants will be assigned to each cohort according to their eligibility.

The first 40 Participants in Part 2 Cohort 1 and Cohort 2 will be randomized to one of two dose levels (RP2D and RP2D-1, Q3W) for dose optimization in a 1:1 ratio to:

- Arm 1: RP2D (as determined in Part 1)
- Arm 2: RP2D -1 (lower dose, as defined by the IRC post Part 1 of the trial based on available clinical data)

No randomization is planned for any other cohort.

### 1.2.2 Central Review

An external ILD Adjudication Committee will retrospectively review all cases of potential ILD/pneumonitis. To ensure adequate evaluation, relevant additional data from within the clinical database and other sources, including imaging data, may be provided to the adjudication committee to fully characterize medical history (e.g., smoking, radiation and pulmonary history), diagnostic evaluation, treatment and outcome of the event. The ILD Adjudication Committee will not influence trial treatment decisions. Further details can be found in the ILD Adjudication Charter.

### 1.2.3 Data Monitoring

A trial IRC will be established to provide medical oversight of participant safety during the conduct of this trial, with a focus on guidance, management of emergent safety issues, and decision making as outlined in the IRC Charter. This includes periodic in-depth review of safety data by trial participant, cohort, and cumulatively, in order to confirm mechanism of action, identify potential off-target toxicities, and to understand the IMP's safety profile and feasibility for further clinical development. The IRC is also a forum for the discussion of other data which could impact the IMP benefit-risk assessment, thereby allowing the IRC to periodically assess the overall benefit-risk of the IMP.

The IRC will be constituted and act according to procedures described in the IRC Charter. The IRC will prepare written minutes of its meetings.

## 2 STATISTICAL HYPOTHESES AND MULTIPLICITY ADJUSTMENT

There is no formal hypothesis testing for this trial.

No formal interim statistical analysis is planned. However, the safety and efficacy data will be monitored on an ongoing basis.

For Part 1, safety data will be continuously monitored for each cohort. The dose levels moving to Part 2 will be determined based on the totality of the data.

Part 2 will further evaluate the anti-tumor activity and safety profile of the IMPs. Based on the totality of the available data (e.g., safety, efficacy, PK etc.), the optimal dose level will be selected for Cohort 1 and Cohort 2 target indication. For each of the Cohorts 3-7, an early futility check on ORR results will be performed after the first 10 participants become response evaluable (see definition in Section 3) in the corresponding cohort. The observed ORR (unconfirmed) results from each cohort of Part 2 Cohort 3 to 7 will be compared to the corresponding target ORR presented in the Table 3. A cohort will be considered futile and stopped for further enrolment if the posterior probability of ORR (unconfirmed) greater than the target value is lower than or equal to 5%. A non-informative Beta (0.001, 0.001) prior will be applied. The false stopping probability was defined as the probability of not meeting the positivity of the futility analysis when the true ORR equals to the value listed in the Table 3. Enrollment will not pause when conducting the analysis.

**Table 3: Criteria of passing the futility analysis in Part 2 Cohorts 3-7**

| Cohort | Indication                                | Target ORR | Min number of responders (unconfirmed) to pass the futility analysis | True ORR | False stopping probability |
|--------|-------------------------------------------|------------|----------------------------------------------------------------------|----------|----------------------------|
| 3      | non-sq NSCLC, AGA-, 2L, post-chemo +/- IO | 20%        | $\geq 1/10$                                                          | 20%      | 10.5%                      |
|        |                                           |            |                                                                      | 25%      | 5.2%                       |
| 4      | sq-NSCLC, AGA-, 1L                        | 50%        | $\geq 3/10$                                                          | 50%      | 5.6%                       |
|        |                                           |            |                                                                      | 55%      | 2.7%                       |
| 5      | sq-NSCLC, AGA-, 2L, post-chemo +/- IO     | 20%        | $\geq 1/10$                                                          | 20%      | 10.5%                      |
|        |                                           |            |                                                                      | 25%      | 5.2%                       |
| 6      |                                           | 40%        | $\geq 2/10$                                                          | 40%      | 4.8%                       |

| Cohort | Indication             | Target ORR | Min number of responders (unconfirmed) to pass the futility analysis | True ORR | False stopping probability |
|--------|------------------------|------------|----------------------------------------------------------------------|----------|----------------------------|
|        | non-sq, AGA+, post-TKI |            |                                                                      | 45%      | 2.2%                       |
| 7      | ES-SCLC, 1L            | 70%        | ≥5/10                                                                | 70%      | 4.5%                       |
|        |                        |            |                                                                      | 75%      | 1.8%                       |

In addition, an expansion evaluation analysis is planned for each of the Cohorts 3-7 after the first 20 participants become response evaluable in each of the cohorts. To facilitate the expansion decision-making, the observed ORR (unconfirmed) results from each cohort will be compared to the lower and higher target ORR for each cohort. A cohort will be stopped for further enrolment if the posterior probability of ORR (unconfirmed) greater than the higher target value is lower than or equal to 5%. A cohort will be considered promising for further enrolment if the posterior probability of ORR (unconfirmed) greater than the lower and higher target values are greater than 80% and 5%, respectively. The lower and higher target ORR values and the corresponding decision criteria for each cohort are listed in [Table 4](#). A non-informative Beta (0.001, 0.001) prior will be applied. In addition, depending on the totality of the data, selected cohorts may be chosen to enroll up to 40 participants per cohort. Of note, these criteria are non-binding and the totality of the available data (e.g., safety, efficacy, PK, etc.) will be reviewed by the sponsor to determine expansion or not. This potential expansion may utilize dosing at the RP2D or RP2D-1. Enrolment for other ongoing cohorts will not pause when conducting the analysis in each cohort.

**Table 4: The criteria in Part 2 Cohorts 3-7 for the expansion decision making**

| Cohort | Indication                             | Lower Target ORR | Higher Target ORR | Not Expansion Criterion N=20 | Consider N=20                             | Expansion Criterion N=20 |
|--------|----------------------------------------|------------------|-------------------|------------------------------|-------------------------------------------|--------------------------|
| 3      | NSQ NSCLC, AGA-, 2L, post chemo +/- IO | 20%              | 35%               | ≤15% (3/20)                  | 20% (4/20) or 25% (5/20)                  | ≥ 30% (6/20)             |
| 4      | SQ, AGA-, 1L                           | 50%              | 60%               | ≤40% (8/20)                  | 45% (9/20) or 50% (10/20) or 55% (11/20)  | ≥ 60% (12/20)            |
| 5      | SQ, AGA-, 2L, post-chemo +/- IO        | 20%              | 30%               | ≤15% (3/20)                  | 20% (4/20) or 25% (5/20)                  | ≥ 30% (6/20)             |
| 6      | NSQ, AGA+, post-TKI                    | 40%              | 50%               | ≤30% (6/20)                  | 35% (7/20) or 40% (8/20) or 45% (9/20)    | ≥ 40% (10/20)            |
| 7      | ES-SCLC, 1L                            | 70%              | 80%               | ≤60% (12/20)                 | 65% (13/20) or 70% (14/20) or 75% (15/20) | ≥ 80% (16/20)            |

The primary analysis of the trial will be performed when all participants have either discontinued the trial or have been (or have had the chance to be) followed for at least 18 months from the start of IMP, whichever occurs first.



### 3 ANALYSIS SETS

#### Definition of Analysis Sets

The following populations are defined for this study:

| Population                                      | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screened Set                                    | All participants who have signed and dated the trial informed consent form.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Enrolled Set                                    | All participants who were enrolled or randomized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ITT Set                                         | All participants who were randomized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Full Analysis Set (FAS)                         | All trial participants who received (i.e., at least one dose of) IMP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Response Evaluable Set                          | All trial participants in the FAS who have had at least two post-baseline evaluable response assessments, or have PD per RECIST 1.1, or have discontinued from trial treatment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Dose-Limiting Toxicity Evaluation Set (DLT Set) | All trial participants from the FAS who either have completed the DLT evaluation period (Cycle 1) and meet the minimum exposure criterion or have experienced a DLT during the DLT evaluation period.<br><br>Trial participants who do not experience any DLT during the DLT evaluation period are considered to be evaluable if they have been observed for minimum 21 days following the first dose and are considered to have sufficient safety data to conclude that a DLT did not occur. A trial participant is considered to have met the minimum exposure criterion if at least 85% of the intended dose of both IMPs was administered.<br><br>Participants who were enrolled in Part 1 backfill and Part 2 are excluded. |
| PK Analysis Set (PKS)                           | All trial participants enrolled and who received at least one dose of BNT324 or BNT327, and who had at least one measurable blood concentration of BNT327, BNT324 ADC, total anti-B7-H3 antibody, or unconjugated payload P1021.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Immunogenicity Analysis Set                     | All trial participants enrolled and received at least one dose of BNT324 or BNT327, and with at least one valid ADA result post-first dose against BNT324 or BNT327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

PK = pharmacokinetic.

The DLT evaluation set will be used for the evaluation of DLTs in order to support the dose escalation/de-escalation decision per BOIN design. The FAS will be used for all other safety analyses and efficacy analyses. The Response Evaluable Set will only be used for interim efficacy monitoring. The ITT set may be used for efficacy sensitivity analysis in Part 2 Cohort 1 and Cohort 2. PKS will be used for PK analysis. The Immunogenicity Analysis Set will be used for immunogenicity analysis.

ITT set analysis will be based on the planned treatment. All other analyses will be based on the treatment actually taken by the trial participant ('as treated').

#### 3.1 Violations and Deviations

Protocol deviations are failures to adhere to the inclusion/exclusion criteria and protocol requirements and will be classified as important or not important protocol deviations. participant. Important protocol deviations will be defined in the Protocol Deviations Management Plan.

Important protocol deviations will be identified and categorized by medical review periodically before the database lock for the final analysis.

## 4 STATISTICAL ANALYSIS

Statistical analysis will be performed by BioNTech SE Global Biometric Sciences group. All statistical analyses will be carried out using SAS®, Version 9.4 or higher, and/or other statistical software as required.

Any deviations from the planned analyses described in this SAP will be described and justified in the Clinical Trial Report (CTR). This section describes the planned statistical analyses of the efficacy, safety and other endpoints including primary and key secondary endpoints.

### 4.1 General Considerations

Unless otherwise specified, the analyses will be performed separately for Phase Ib (Part 1) by dose level, and Phase II (Part 2) by cohort, dose level, and treatment arm as applicable. Tables, figures and listings will be provided separately for Part 1 and Part 2. Cohort-specific tables and figures may be provided as appropriate. Pooled safety analysis may be performed for selected dose levels across Part 1 and Part 2. Pooled efficacy analysis for selected tumor types may also be performed by dose level across Part 1 and Part 2 as appropriate. For interim cohort-specific futility checks and expansion decision making analyses in Part 2, analyses on selected cohorts will be performed.

Unless specified otherwise, each trial part will be summarized and analyzed separately. TFLs will be presented by trial part, cohort, dose level, visit, and timepoint as applicable.

All medical history, prior and concomitant medications, prior and concomitant procedures, prior and subsequent anti-cancer therapies, and adverse events will be coded using MedDRA; the primary system organ class (SOC) will be used for analyses. All medications will be coded using the WHO drug dictionary. The version of the dictionaries will be the dictionary current at the time of database lock/data snapshot.

Except where specified otherwise, all continuous variables will be summarized with descriptive statistics (the number of non-missing values, mean, standard deviation (SD), median, lower quartile (Q1), upper quartile (Q3), minimum and maximum) and all categorical variables will be summarized with frequency counts and percentages in each category, by trial part/dose level/cohort/treatment arm, as appropriate.

Overall summaries may be provided. For reporting summary statistics, the mean and median will be displayed to one decimal place greater than the original value, and the measure of variability (e.g., standard deviation) will be displayed to two decimal places greater than the original value; a maximum of four significant digits will be displayed.

Time-to-event-endpoints will be analyzed using Kaplan-Meier methodology by treatment arm/cohort as appropriate and censored in accordance with [FDA Guidance](#). Censoring rules are defined in [Appendix 3](#).

The median survival time (including two-sided 95% confidence limits according to [Brookmeyer, R., and Crowley, J. \(1982\)](#) with log-log transformation and the first and third quartile will be presented for each treatment group/cohort. Survival rates (including two-sided 95% confidence interval based on Greenwood's formula [[Greenwood, M. \(1926\)](#) ]) as well as the number and percentage of trial participants with events, censored and under risk will be displayed for selected timepoints (e.g., at 3, 6, 12 months).

The time-to-event analysis will be illustrated using Kaplan-Meier plots.



Data up to and including the clinical database cut-off date for the statistical analysis will be included in the statistical analysis. For OS analysis, a survival sweep will be performed and any data collected after the cut-off date (but before the database lock) will be used to determine if a participant was alive at the cut-off date.

For conversion of days to years, months, or weeks, the following conversion factors will be used:

- 1 year = 365.25 days
- 1 month =  $365.25/12 = 30.4375$  days
- 1 week =  $365.25/52 = 7.024$  days

#### 4.1.1 Definitions, Derivations and Classifications

##### Definition of Baseline

The baseline value for all safety endpoints (i.e., laboratory parameters and vital signs) will be the last available non-missing value (including unscheduled assessments) prior to the first dose of trial treatment. Assessments taken on the same day as the first dose of trial treatment will be assumed to be before trial treatment unless indicated to be obtained post-the first dose of trial treatment based on available start date/time combination or collected electronic Case Report Form (eCRF) information that identifies the individual event/medication.

The baseline value for all electrocardiogram parameters will be the average of the triplicate measurements from the last available non-missing ECG value prior to the first dose of trial treatment. Assessments taken on the same day as treatment will be assumed to be before BNT324 administration unless indicated to be obtained post based on available start date/time combination or collected electronic Case Report Form (eCRF) information that identifies the individual event/medication. based on available start date/time combination or collected electronic Case Report Form (eCRF) information that identifies the individual event/medication.

The baseline value for all efficacy endpoints will be the last available non-missing value prior to the first dose of trial treatment.

For PRO endpoints, the last available non-missing value prior to first dose of trial treatment will be used as baseline.

##### Definition of Day 1

Study Day 1 is defined as the first day of trial treatment start date. This will also be considered as the reference date.

##### Definition of Study Day

Study Day will be computed as follows:

- Study Day = (Date of event/assessment – reference date) + 1 if the date of the event is on or after the reference date.
- Study Day = (Date of event – reference date) if the date of the event is prior to the reference date.
- Study Day post-treatment: Relative Study Day post treatment = date of interest – date of most recent treatment + 1 where most recent treatment refers to the last treatment received.

## Derivations Involving Dates

The number of days between two dates, the date of event and the reference date, will be calculated as (Date of event – Reference date) + 1 when the event date is on or after the reference date; or, as (Reference date – Date of event) + 1 when the event date is before the reference date. The resulting value is then converted to years, months, or weeks, as appropriate. Partial dates will be imputed before calculating the number of days (See [Appendix 7](#) for partial or unknown date imputation rules).

## Common Calculations

Change from baseline (CHG) will be calculated as:

Change from baseline = Actual (observed) value at post baseline visit – baseline value

Percent change from baseline (PCHG) will be calculated as:

Percent change from baseline (%) =  $100 \times (\text{Change from baseline at post baseline visit} / \text{Baseline value})$

If the baseline is not available, the change from baseline and percent change from baseline will not be calculated and will remain missing.

## Common Analyses

For primary and secondary efficacy binary endpoints, the point estimates and 95% exact binomial confidence intervals (Clopper-Pearson) will be provided.

For time-to-event endpoints, Kaplan-Meier curves (product limit estimates) will be presented by trial part, dose level, cohort, and treatment arm, as appropriate, together with a summary of associated statistics (e.g., median survival time and survival rates including the corresponding two-sided 95% CIs).

## Demographics and Disposition

Age in years at time of enrollment will be derived relative to the date of informed consent (IC) as: (year of consent – year of birth).

Additionally, age will be categorized into two categories: <65 and ≥65 years.

Body mass index (BMI) will be derived as weight in kilograms (kg) divided by height in meters squared (i.e.,  $\text{weight} / [\text{height} \times \text{height}]$ ). Units for BMI are expressed as kg/m<sup>2</sup>. BMI will be categorized into four categories: <18.5, ≥18.5-<25, ≥25-<30, and ≥30 kg/m<sup>2</sup>.

Duration of participant participation will be derived relative to the date of receiving the first dose. The date of the visit corresponding to the final evaluation/study termination, or the data cutoff date if the participant is still on trial, will be used as the end date of participant participation.

If participants have temperature in CRF reported using Fahrenheit, the values reported can be converted programmatically to Celsius using the following formula:  $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$  to standardize units for easier interpretation, if required for reporting.

## Subgroup Definitions

The following additional subgroups will be derived for statistical analysis:

- Age group (<65, ≥65 years),
- Region (Asia-Pacific, North America, Europe, Rest of World),

- Number of prior lines of therapy (1, 2, 3+) in all settings,
- Number of prior lines of therapy (1, 2, 3+) in advanced or metastatic settings,
- Number of prior lines of therapy (1, 2+) in adjuvant or neo-adjuvant settings,
- Smoking status: never, current, former.

#### 4.1.2 Windowing, Unscheduled Visits and Repeat Test Results

For by-visit analysis, the corresponding endpoints will be mapped to the analysis visit according to the nominal visits.

#### 4.1.3 Handling of Intercurrent Events

Expected intercurrent events of interest for this trial are initiation of subsequent systemic anti-cancer treatment for primary disease and discontinuation of trial treatment. Initiation of subsequent radiotherapy for primary disease, except for palliative treatment, if reported, will also be included as an intercurrent event the same way as the systemic anti-cancer treatments will.

The intercurrent event strategies specified in [Table 5](#) below will be used for efficacy endpoints.

**Table 5: Strategies for handling intercurrent events**

| Intercurrent events of interest                                                                                                                                           | Strategy 1 (PFS, ORR, DOR, DCR, TTR) | Strategy 2 (OS)           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------|
| Initiation of subsequent systemic anti-cancer treatment for primary disease, or initiation of subsequent radiotherapy for primary disease except for palliative treatment | Hypothetical strategy                | Treatment policy strategy |
| Discontinuation of trial treatment                                                                                                                                        | Treatment policy strategy            | Treatment policy strategy |

The hypothetical strategy will be used to assess the effect of treatment on RECIST 1.1 based efficacy endpoints (PFS, ORR, DOR, TTR, DCR) in the hypothetical scenario in which initiation of subsequent systemic anti-cancer treatment or initiation of subsequent radiotherapy (except palliative treatment) does not occur. In these analyses, exclusion of possible impact of subsequent systemic anti-cancer treatment and subsequent radiotherapy (except palliative treatment) in delaying/accelerating event of interest will be handled by censoring observations before the start of the subsequent anti-cancer treatment or before the start of subsequent radiotherapy (except palliative treatment) for time to event analyses. For ORR and DCR analyses, if the response from the participants happens after the initiation of subsequent systemic anti-cancer treatment or after the initiation of subsequent radiotherapy (except palliative treatment), the participants will be considered as non-responder. This is because subsequent anti-cancer systemic treatment or subsequent radiotherapy (except palliative treatment) will impact the efficacy of trial treatment. Efficacy assessment after subsequent systemic anti-cancer treatment will be therefore excluded from the analysis.

The treatment policy strategy will be used to assess the treatment effect on RECIST 1.1 based efficacy endpoints (PFS, ORR, DOR, TTR, DCR) and OS irrespective of the trial treatment discontinuation; the data collected for these endpoints will be therefore used

regardless of whether the participant discontinued the trial treatment. For OS analysis, the data collected will be used regardless of initiation of subsequent anti-cancer treatment (treatment policy strategy). This is to reflect clinical practice.

#### 4.1.4 Dropouts, Missing/Incomplete Data, and Outliers

Missing data occurs when any requested data is not provided, leading to blank fields on the collection instrument. These data will be indicated using a “blank” in participant listing displays. Answers such as “Not applicable” and “Not evaluable” are not considered to be missing data and should be displayed as such.

Missing efficacy assessments for PFS, OS, and DOR analyses will be handled according to the censoring rules defined in [Appendix 3](#).

Missing or incomplete dates will be imputed according to the rules specified in [Appendix 7](#).

Uncoded events and procedures at the time of analysis will be assigned the string “UNCODED” as the system organ class, and the reported term will be used as the preferred term so they can be included in the summary tables. In the final dataset, all the adverse events and procedures should be coded.

Uncoded medications at the time of analysis will be assigned the string “UNCODED” as the Anatomical-Therapeutic-Chemical (ATC) summary level(s) and the verbatim term will be used as the preferred name, so they can be included in the summary tables. In the final dataset, all the medications will be coded.

## 4.2 Participant Disposition

The number and percentage of participants who have signed informed consent (screened), are enrolled or randomized (Part 2 Cohort 1 and 2), received each trial treatment, discontinued from the trial treatment, and discontinued from the trial will be tabulated by trial part, dose level, cohort, treatment arm, and overall, as appropriate. Reasons for discontinuation of treatment and the trial will also be tabulated by trial part, dose level, cohort, treatment arm, and overall, as appropriate, using numbers and percentages. For those participants who are considered to have failed screening, the reason(s) for failure will be tabulated. Participants who were re-screened and failed will be counted only once in the summary table under the relevant category.

Important protocol deviations will be summarized by deviation category for each trial part, dose level, cohort, treatment arm, and overall, as appropriate for the FAS; The listing will include an indicator for important or not important.

The number and percentage of participants included and excluded from the analysis sets will be tabulated for each trial part, dose level, cohort, treatment arm, and overall, as appropriate. Reason(s) for exclusion from each set will be summarized overall and by reason for exclusion from the Enrolled Set.

A descriptive summary of the duration of the participant participation in the study will be produced by trial part, dose level, cohort, and treatment arm, as appropriate, for the FAS.

The duration of being on treatment and ongoing study is flexible for participants in this trial therefore the number of participants completed the treatment and number of participants completed the trial will not be reported.

By-participant data listings will present supporting data for participant disposition for the Enrolled Set.

### 4.3 Participant Characteristics

Participant characteristics include demographics, baseline characteristics and assessments, medical history, and information related to past medications, therapies and procedures. All participant characteristics will be based on the specified analysis set(s) and will be summarized by trial part, dose level, cohort, as appropriate, for the FAS.

#### 4.3.1 Demographic and Baseline Characteristics

Descriptive statistics will be used to summarize demographic and baseline characteristics. Demographic data include:

- age (years),
- Age group (<65, ≥65 years),
- Sex (female [including childbearing potential]),
- Race,
- Ethnicity,
- Region (Asia, US/Canada, Rest of World),

Baseline characteristics include:

- weight (kg),
- height (cm),
- BMI group (<18.5, ≥18.5 to <25, ≥25 to <30, ≥30),
- Smoking status (never, current, and former) and substance (cigarettes, cigars, other tobacco use, and other marijuana use),
- Alcohol status (never, current, and former) and type (beer, wine, spirits, and other),
- Number of prior lines of therapy (1, 2, 3+) for advanced/metastatic disease,
- Received prior therapy in neo-adjuvant setting (yes vs. no),
- Received prior therapy in adjuvant setting (yes vs. no),
- Received prior anti-cancer surgical procedure (yes vs. no),
- Received prior radiotherapy (yes vs. no),
- Time from completion of prior therapy for earlier lung cancer stages (i.e. received prior systemic anti-cancer therapy in neoadjuvant, adjuvant, or locally advanced treatment setting) to start of IMPs as the 1L anticancer therapy in Stage III or IV per AJCC in the “Cancer Characteristic [lung]” eCRF (≥6 months .vs <6 months).

Additional demographic and baseline characteristics may be summarized as appropriate.

Summaries will be produced for the FAS based on the data collected in the “Demographics”, “Alcohol Substance Use”, “Smoking Substance Use”, “Physical Measurements”, and “Prior Systemic Therapies for Primary Disease” CRFs.

By-participant listings will present supporting data for the FAS.

#### 4.3.2 Baseline Assessments

Descriptive statistics will be used to summarize the following baseline assessments: safety endpoints, efficacy assessments, laboratory parameters, vital signs, ECG parameters, and HRQoL endpoints (Part 2 Cohort 1 and 2 only).

Results from the baseline assessments will be used for the analysis to investigate the change in assessment outcomes over time.

By-participant data listings will be produced for the FAS.

#### 4.3.3 Medical History

Medical history will be classified using MedDRA terminology. Medical history will be summarized using numbers and percentages by primary system organ class, and preferred term, for the FAS.

Medical History conditions/surgeries are defined as those conditions/surgeries which stop prior to enrollment. Current medical conditions are defined as those conditions which started prior to enrollment and are ongoing/stopped after enrollment. Medical conditions/surgeries data captured on the 'Medical History' page of the CRF and will be summarized using numbers and percentages for the FAS.

By-participant data listings will be produced for the FAS.

#### 4.3.4 Disease-Specific Characteristics

Descriptive statistics will be used to summarize disease history. Cancer history will include

- Primary type of cancer at initial diagnosis and study entry time from initial diagnosis to date of first dose, the TNM stage at initial diagnosis, the TNM stage at study entry, time from most recent progression to start of any trial treatment, and site of metastasis as recorded in the "Cancer Diagnosis and Staging [Solid Tumor] CRF.
- Primary histopathology at initial diagnosis and study entry, location and laterality of primary tumor, histologic grade at initial diagnosis, whether primary tumor is present at study entry, extent of disease as assessed Veterans Administration Lung Study Group, as recorded in the "Cancer Characteristics [Lung]" CRF.
- AGA status at the study entry (i.e., + vs.-) for NSCLC participants, where AGA+ is defined as actionable oncogenic alteration: EGFR exon 19 deletion or exon 21 L858R mutation positive, EGFR S768I, L861Q, and/or G719X mutation positive, EGFR exon 20 insertion mutation positive, KRAS G12C mutation positive, ROS1 rearrangement positive, BRAF V600E mutation positive, NTRK1/2/3 gene fusion positive, METex14 skipping mutation positive, RET rearrangement positive, ERBB2 (HER2) mutation positive, ALK rearrangement positive, NRG1 gene fusion positive as recorded in the "Cancer Molecular Profile [Lung]" CRF. Any other mutation outside the list is considered AGA-.
- Number of lines of prior systemic therapy recorded in the "Prior Systemic Therapies for Primary Disease" CRF.
- ECOG performance status at study entry as recorded in the "ECOG" CRF.

Additionally, the number of metastatic sites at enrollment will be summarized using descriptive statistics. The metastatic site(s) involved at enrollment will also be tabulated



using numbers and percentages. The data will be summarized as recorded in the “Cancer Diagnosis and Staging [Solid Tumor]” CRF.

The sum of diameters for the target lesions measured at baseline will be summarized by using descriptive statistics. The data will be summarized as recorded by the investigator on the “Tumor Assessment – Target Lesion (Baseline)” CRF.

Additional baseline disease-specific characteristics may also be summarized as appropriate.

By-participant listings will be provided for the FAS.

#### **4.3.5 Medications, Therapies and Procedures**

##### **4.3.5.1 Prior and Concomitant Medications**

Prior medications are defined as any medications and therapies with a start and end date before the first dose of trial treatment.

Concomitant medications are defined as follows:

- Any medication that started before the first dose of trial treatment and was ongoing at the time of the first dose of trial treatment or ended on the date of the first dose of trial treatment.
- Any medication that with a start date on or after the date of the first dose of trial treatment and up to 90 days after the last dose of trial treatment or the initiation of new anti-cancer surgery, whichever occurs first.

See [Appendix 7](#) for handling of partial dates for medications. If a medication or therapy cannot be classified due to completely missing dates, it will be classified as concomitant.

Medications with a start date more than 90 days after the last dose of trial treatment or medications with a start date after the initiation of new anti-cancer systemic therapy will be classified as post-treatment medications.

Prior and concomitant therapies/medications will be summarized by ATC therapeutic class (ATC Level 2), ATC pharmacological class (ATC Level 3), and chemical substance (ATC Level 5)/preferred name for the FAS. The data of prior and concomitant medications will be summarized as recorded in the “Prior and Concomitant Medications” CRF.

All prior, concomitant medications, and post-treatment medications will be listed for the FAS.

##### **4.3.5.2 Disease-Specific Medications and Procedures**

Prior anti-cancer systemic therapies, prior radiotherapies and prior surgical procedures for the diseases under study will be summarized by trial part, dose level, cohort, treatment arm and overall using descriptive statistics for the FAS, as appropriate. The data of whether any prior systemic therapy, prior radiotherapy, or surgical procedure was performed will be recorded in the “Prior Anti-Cancer Therapies Evaluation” CRF.

Descriptive summaries of prior systemic therapies include the number of prior lines of therapy, the setting of the most recent line of systemic therapy, treatment type of the most recent line of systemic therapy, the best response to the most recent line of systemic therapy; the reason for discontinuation of the last systemic therapy, and the time from



when the last systemic therapy ended to study enrollment. The data will be summarized as recorded in the “Prior Systemic Therapies for Primary Disease” CRF.

Descriptive summaries of prior radiotherapies include the location and laterality, the treatment intent and the procedure used to administer the last radiotherapy. The data will be summarized as recorded in the “Prior Radiotherapy for Primary Disease” CRF.

Descriptive summaries of prior surgical procedures include whether any surgical procedures were performed, the type of procedure, location and laterality of the procedure, and the residual disease remaining after surgery. The data will be summarized as recorded in the “Prior Surgical Procedures for Primary Disease” CRF.

In addition, subsequent anti-cancer systemic therapies, radiotherapies and surgical procedures for the diseases under study will be summarized for the FAS using descriptive statistics. The data of whether any post-study treatment systemic therapy, post-study treatment prior radiotherapy, or post-study treatment surgical procedure was performed will be recorded in the “Subsequent Anti-Cancer Therapies Evaluation” CRF. Descriptive summaries of subsequent anti-cancer therapies include the intent of the radiotherapy; intent and procedure type for surgeries; and the number of systemic therapies. Systemic therapies/medications will also be summarized by ATC therapeutic class (ATC Level 2), ATC pharmacological class (ATC Level 3), and chemical substance (ATC Level 5)/preferred name for the FAS.

By-participant data listings will be produced for the FAS as appropriate.

#### **4.3.5.3 Prior and Concomitant Procedures**

Other non-oncological prior and concomitant procedures will be summarized by system organ class and preferred term for FAS. The data will be summarized as recorded in the “Medical History” CRF for procedure started before or after the participants enrolled in the study.

Prior procedures/non-drug therapies are defined as follows:

- Any procedure/non-drug therapy with a start and end date before the first dose of trial treatment.

Concomitant procedures/non-drug therapies are defined as follows:

- Any procedure/non-drug therapy that started before the first dose of trial treatment and was ongoing at the time of the first dose of trial treatment or ended on the date of the first dose of trial treatment.
- Any procedure/non-drug therapy with a start date on or after the date of the first dose of trial treatment and up to 90 days after the last dose of trial treatment or the initiation of new anti-cancer surgery, whichever occurs first.

Procedures/non-drug therapies with a start date more than 90 days after the last dose of trial treatment will be classified as post-treatment procedures/non-drug therapies.

By-participant data listings will be produced for the FAS.

## 4.4 Primary Endpoint(s) Analyses

### 4.4.1 Primary Endpoint(s) Analysis – Part 1

The primary objective of Part 1 of the study is to determine the RP2D of BNT324 in combination with BNT327.

#### 4.4.1.1 Definition of Endpoint(s)

The primary endpoints of Part 1 of the study include

- Occurrence of DLTs during the DLT evaluation period (Cycle 1, 21 days).
- Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.
- Occurrence of drug interruption, reduction, and treatment discontinuations due to TEAEs by dose level from the time of the first dose of IMP to 90 days after the last dose of IMP or until new anticancer therapy is started, whichever occurs first

The DLT criteria are defined in the protocol Section 6.3.1.3. The data will be summarized as recorded in the “Adverse Events” CRF.

A TEAE is defined as any AE with an onset date on or after the first administration of IMP (if the AE was absent before the first administration of IMP) or worsened after the first administration of IMP (if the AE was present before the first administration of IMP).

Note: AEs with an onset more than 90 days after the last administration of IMP or up to 10 days prior to the start of new anticancer therapy will only be considered treatment-emergent if assessed as related to the IMP by the investigator.

#### 4.4.1.2 Main Analytical Approach

No formal statistical comparisons between treatment arms will be made in this trial.

The primary analyses will be performed using the FAS.

##### *Dose-limiting toxicities (DLT)*

DLT analysis will be performed in Part I only, excluding the backfill participants. The number and percentage of trial participants reporting at least one DLT will be summarized by dose level.

DLTs will also be presented in terms of listings presenting the reported term and MedDRA PT and SOC term, its time of onset, duration, and outcome, relationship, NCI CTCAE grade, and seriousness including dose exposure data.

The estimations of the DLT rate with corresponding 90% CIs at each dose level based on isotonic regression will be provided. The confidence interval will be calculated based on the Bayesian posterior credible interval with a prior distribution of Beta (0.05, 0.05).

##### *Treatment-emergent adverse events (TEAEs)*

AEs will be graded according to NCI CTCAE version 5.0 and coded using the most recent version of the MedDRA coding.

See CTP v3.1 Section 10.4.1 for definition of AEs and TEAEs, Section 10.4.2 for definitions of SAEs., and Section 8.5.8 for definitions of AESIs.

TEAEs will be summarized overall and by dose level.

The number and percentage of trial participants reporting at least one TEAE will be summarized by PT nested within the SOC for each of the following AE types:

- Any TEAEs
- Drug-related TEAEs
- Grade  $\geq 3$  TEAEs
- Any serious TEAEs

The number and percentage of trial participants reporting at least one TEAE will be summarized by PT for each of the following AE types:

- Procedure-related TEAEs
- Drug-related Grade  $\geq 3$  TEAEs
- Drug-related serious TEAE
- Serious TEAE leading to death
- Drug-related serious TEAE leading to death
- TEAEs leading to dose reduction
- TEAEs leading to drug interruption
- TEAEs leading to drug delay
- TEAEs leading to permanent discontinuation of treatment
- Drug-related TEAEs leading to dose reduction
- Drug-related TEAEs leading to drug interruption
- Drug-related TEAEs leading to drug delay
- Drug-related TEAEs leading to permanent discontinuation of treatment
- Treatment emergent AESIs
- Drug-related treatment emergent AESIs

The number and percentage of trial participants reporting at least one TEAE will be summarized by PT nested within the SOC (or AESI category) for each of the above TEAE types. Additional TEAE tables at specific incidence rate cut-offs by PT by decreasing incidence in the total columns may be provided for above TEAE types as appropriate.

Drug-related TEAEs will be split (where applicable) by BNT324 only, BNT327 only, BNT324 or BNT327 (if TEAE was related to either drug), BNT324 and BNT327 (if TEAE was related to both drugs). TEAEs leading to dose reduction, drug interruption, drug delay, and drug discontinuation will be presented in the same manner. For drug-related TEAE leading to dose reduction, drug interruption, drug delay, and drug discontinuation, TEAEs that are considered related to either BNT324 or BNT327 will be included.

Moreover, the number and percentage of trial participants with any TEAE, drug-related TEAE, procedure-related TEAE, serious TEAE, and drug-related serious TEAE will be summarized by the worst toxicity grade by PT nested within the SOC.

#### **4.4.1.3 Sensitivity Analyses**

Not planned.

#### **4.4.1.4      Supplementary Analyses**

Not planned.

#### **4.4.1.5      Multiplicity Adjustment**

There is no multiplicity adjustment. No formal statistical comparisons between treatment arms will be made in this trial.

#### **4.4.1.6      Subgroup Analyses for Primary Endpoint**

Subgroup analyses for safety endpoints are not planned.

### **4.4.2      Primary Endpoint(s) Analysis – Part 2**

The primary objectives of Part 2 of the study include

- To determine the optimal dose of BNT324 in combination with BNT327 by assessing the safety profile and efficacy of the combination therapy in the randomized dose optimization cohorts in the lead cohorts (Cohort 1 treatment naive non-sq NSCLC and Cohort 2 relapsed/progressive SCLC).
- To evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1 in Cohorts 3-7.

#### **4.4.2.1      Definition of Endpoint(s)**

For Cohort 1 and 2, the primary endpoints of Part 2 of the study include

- Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.
- Occurrence of drug interruption, reduction, and treatment discontinuations due to TEAEs by dose level from the time of the first dose of IMP to 90 days after the last dose of IMP or until new anticancer therapy is started, whichever occurs first

For all cohorts in Part 2, the primary endpoint includes

- ORR, defined as the proportion of participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment).

The definition of TEAE is provided in Section [4.4.1.1](#). The new anticancer therapy doesn't include new anticancer palliative radiotherapy and new anticancer palliative, diagnostic, and supportive care surgery.

ORR is defined as the proportion of trial participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment). Trial participants not meeting the criteria for CR or PR, including those without any post-baseline tumor assessments, will be considered as non-responders. The derivation rules for confirmed response are provided in [Appendix 4](#).

#### **4.4.2.2      Main Analytical Approach**

No formal statistical comparisons between treatment arms will be made in this trial.

The primary analyses will be performed using the FAS.

The analysis of safety endpoints will follow the same approach as stated in Section [4.4.1.2](#).

ORR will be summarized with absolute and relative frequencies along with two-sided 90% Clopper-Pearson CIs by cohort and treatment arm if applicable.

#### **4.4.2.3 Sensitivity Analyses**

For Part 2 Cohort 1 and Cohort 2, ORR analysis based on the ITT set and planned treatment may be performed as sensitivity analysis.

The same analysis method for the confirmed ORR in Section [4.4.2.2](#) will be applied.

#### **4.4.2.4 Supplementary Analyses**

Not planned.

#### **4.4.2.5 Multiplicity Adjustment**

There is no multiplicity adjustment. No formal statistical comparisons between treatment arms will be made in this trial.

#### **4.4.2.6 Subgroup Analyses for Primary Endpoint**

The generalizability of ORR results may be investigated by estimating the treatment effect in subgroups based on key baseline demographics, disease characteristics, etc. (see Section [4.1.1](#) for details). Additional details related to subgroup analyses are specified in Section [4.9](#).

### **4.5 Secondary Endpoint(s) Analyses**

#### **4.5.1 Secondary Endpoint(s) Analyses – Part 1**

The secondary objectives of Part 1 of the study are to evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1. The corresponding endpoints include

- ORR, defined as the proportion of participants in whom a confirmed CR or PR is observed as best overall response (per RECIST 1.1 based on the investigator's assessment).
- DCR, defined as the proportion of participants with confirmed CR, PR, or SD as best overall response (per RECIST 1.1 based on the investigator's assessment).

##### **4.5.1.1 Definition of Endpoint(s)**

The definition of ORR and DCR are provided in Section [4.5.1](#).

The derivation rules for BOR are provided in [Appendix 4](#)

##### **4.5.1.2 Main Analytical Approach**

The number and percentages of participants with each category of best response will be summarized by dose level and tumor type as appropriate, along with ORR and DCR. Estimates of rates, along with the corresponding two-sided Clopper-Pearson 90% CIs, will be provided for overall and by dose level.

#### 4.5.1.3 Sensitivity Analyses

For ORR and DCR sensitivity analyses, participants with unconfirmed responses will be included in the analyses using the same analysis methods described in Section 4.4.2.2.

For Part 2 Cohort 1 and Cohort 2, the secondary endpoints analyses based on the ITT set and planned treatment may be performed as sensitivity analyses.

The same analysis method in Section 4.5.1.2 will be applied.

#### 4.5.1.4 Supplementary Analyses

Not planned.

### 4.5.2 Secondary Endpoint(s) Analyses – Part 2

The secondary objectives of Part 2 of the study include

1. To evaluate the efficacy (other than ORR) of BNT324 in combination with BNT327 in dose optimization/signal seeking cohorts 1-7.
2. To assess the safety of BNT324 in combination with BNT327 in the signal seeking Cohorts 3-7.

#### 4.5.2.1 Secondary Endpoint 1 Analyses

The corresponding endpoints for the secondary objective 1 include

- PFS defined as the time from first dose of IMP to the first objective tumor progression (PD) or death from any cause, whichever occurs first, per RECIST 1.1 based on the investigator's assessment.
- DOR, defined as the time from first objective response (CR or PR) to first occurrence of objective tumor progression (PD) or death from any cause, whichever occurs first (per RECIST 1.1 based on the investigator's assessment).
- OS, defined as the time from first dose of IMP to death from any cause.
- DCR, defined as the proportion of participants with confirmed CR, PR, or SD as best overall response (per RECIST 1.1 based on the investigator's assessment).
- TTR defined as the time from first dose of IMP to first objective response (CR or PR per RECIST 1.1 based on the investigator's assessment).

##### 4.5.2.1.1 Definition of endpoint

PFS assessed by the investigator is defined as the time from the date of first dose of trial treatment to the first objective tumor progression (per RECIST 1.1) or death from any cause, whichever occurs first. PFS time is calculated as PFS event date minus the date of first dose of trial treatment, plus 1 and converted to months (i.e.,  $[\text{PFS event/censor date} - \text{the date of first dose of trial treatment} + 1]/30.4375$ ).

OS is defined as the time from the date of first dose of trial treatment to death due to any cause. OS time is calculated as the death date minus the date of first dose of trial treatment, plus 1 and converted to months (i.e.,  $[\text{Death/Censor Date} - \text{the date of first dose of trial treatment} + 1]/30.4375$ ).



The definitions of DOR, DCR, and TTR are provided in the Section 4.5.2.1. DOR is calculated as the progression/death date minus the date of first CR or PR, plus 1. The censoring rules for DOR are described in Appendix 3.

#### 4.5.2.1.2 Main Analytical Approach

##### PFS

PFS will be summarized using Kaplan-Meier methodology by trial arm and censored in accordance with the Food and Drug Administration (US) (FDA) Guidance: “[Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics](#)”. Censoring rules are defined in Appendix 3.

Additionally, Kaplan-Meier curves (product limit estimates) will be presented by cohort and treatment arms if applicable, together with a summary of associated statistics (e.g., median survival time and survival rates including the corresponding two-sided 95% CI).

The median PFS time (including two-sided 95% CI) and the first and third quartile will be presented for each cohort and treatment arm if applicable. PFS rates (including two-sided 95% confidence interval) as well as the number and percentage of participants with events, censored and under risk will be displayed for selected timepoints (e.g., at 3, 6, 9, 12 months).

The number of participants with PFS and censoring events (along with reasons for censoring) will also be summarized.

By-participant data listings will be produced.

##### OS

OS will be analyzed with the same methods as PFS defined above. Participants who are alive will be censored at the last known alive date. Detailed censoring rules for OS are defined in Table 6. Summaries of the number and percentage of participants who have died, and were censored will be also provided.

**Table 6: Censoring algorithm for overall survival**

| Situation                                                                          | Date participant has event or is censored    | Censored |
|------------------------------------------------------------------------------------|----------------------------------------------|----------|
| Death                                                                              | Date of death                                | No       |
| Participants confirmed alive during the trial                                      | Date of last contact with known alive status | Yes      |
| Participant lost to follow-up                                                      | Date of last contact with known alive status | Yes      |
| Participant withdrew consent from trial                                            | Date of withdrawal of consent                | Yes      |
| Known to be alive or died after database cutoff date and before database lock date | Date of database cutoff                      | Yes      |

##### DOR

DOR will be analysed descriptively by cohort and treatment arm if applicable with duration as a continuous variable. Kaplan-Meier plots, median. DOR times (including two-sided 95% CIs), and the first and third quartile will be presented for each cohort and treatment arm. DOR rates (including two-sided 95% CIs) as well as the number and percentage of participants with failure, censored, and under risk will be displayed for selected timepoints, e.g., at 3, 6, 9, 12 months.



### DCR

DCR will be analyzed with the same method as Section 4.5.1.2.

### TTR

TTR will be summarized by mean, standard deviation, median, minimum value and maximum value. TTR is only for participants whose confirmed BOR is CR or PR.

#### **4.5.2.1.3 Sensitivity Analyses**

For ORR and DCR sensitivity analyses, participants with unconfirmed responses will be included in the analyses using the same analysis methods described in Section 4.5.2.1.2.

For Part 2 Cohort 1 and Cohort 2, the secondary efficacy endpoints analyses based on the ITT set and planned treatment may be performed as sensitivity analyses.

The same analysis method in Section 4.5.2.1.2 will be applied. For participants who were randomized did not receive the IMP, the corresponding starting time for PFS, OS, and TTR are defined as the time of randomization.

#### **4.5.2.1.4 Supplementary Analyses**

Not planned.

#### **4.5.2.2 Secondary Endpoint 2 Analyses**

The corresponding endpoints for the objective 2 include

- Occurrence of TEAEs, serious TEAEs, treatment-related TEAEs, and treatment-related serious TEAEs from the time of the first dose of IMPs to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.
- Occurrence of drug interruption, reduction, and treatment discontinuation due to TEAEs from the time of the first dose of IMP to 90 days after the last IMP dose or until new anticancer therapy is started, whichever occurs first.

##### **4.5.2.2.1 Definition of Endpoint(s)**

The definition of TEAE is provided in Section 4.4.2.1.

##### **4.5.2.2.2 Main Analytical Approach**

The safety endpoints will be analyzed with the same method as Section 4.4.2.2 for each of Cohorts 3-7.

##### **4.5.2.2.3 Sensitivity Analyses**

Not planned.

##### **4.5.2.2.4 Supplementary Analyses**

Not planned.

## 4.6 Exploratory Endpoint(s) Analyses

### Part 1 exploratory endpoints analyses – PFS, DOR, OS, and TTR

The exploratory objective for Part 1 is to evaluate the efficacy of BNT324 in combination with BNT327 according to RECIST 1.1.

The corresponding exploratory endpoints include

- PFS defined as the time from first dose of IMP to the first objective tumor progression (PD per RECIST 1.1) or death from any cause, whichever occurs first (based on the investigator's assessment).
- DOR, defined as the time from first objective response (CR or PR) to first occurrence of objective tumor progression (PD) or death from any cause, whichever occurs first (per RECIST 1.1 based on the investigator's assessment).
- OS, defined as the time from first dose of IMP to death from any cause.
- TTR defined as the time from first dose of IMP to first objective response (CR or PR per RECIST 1.1 based on the investigator's assessment).

The exploratory endpoints will be analyzed with the same method as Section 4.5.2.1.2 by dose level, including the backfill participants. To assess the impact of intent-to-treat, a sensitivity analysis of the secondary efficacy endpoints based on the ITT set may be performed.

## 4.7 Other Analyses

Analysis methods for other exploratory endpoints (PK, immunogenicity, biomarker, and PRO endpoints) are provided in this section.

### 4.7.1 Pharmacokinetic Analyses

The PK objective for this trial is to assess the PK concentrations of BNT324 ADC, total anti-B7-H3 antibody, free payload P1021, and BNT327 in serum or plasma.

Serum concentration of BNT324 ADC, total anti-B7H3 antibody, and BNT327, and plasma concentration of free payload P1021 will be listed and summarized using descriptive statistics based on the PKS.

Mean ( $\pm$  standard deviation) serum and plasma concentrations will be plotted over time on linear and semi-logarithmic scales based on the PKS. Individual serum and plasma concentrations may also be plotted.

Serum or plasma concentrations will be summarized using descriptive statistics (number of participants [N], number of observations [n], arithmetic mean, standard deviation, coefficient of variation percentage (CV%), geometric mean, geometric CV%, median, minimum, maximum) by dose level combining Part 1 and Part 2 participants and by primary cancer type at study entry as recorded in the "Cancer Diagnosis and Staging [Solid Tumor]" eCRF, cycle, day and scheduled time points, as appropriate.

Deviations and trial data will be reviewed to identify participants or data points which may be adversely affected and thus may be excluded from the PK analyses as considered necessary by the analyst. Samples collected outside of the time window of the specified time post-dose will be included for the purpose of PK parameter calculation but will be excluded for the purpose of summarizing concentrations of each time point. Changes to the procedures or events which may impact the quality of the PK data will be considered

as deviations from the clinical trial protocol. These changes or events will include any circumstances that will alter the evaluation of the PK. Examples include, but may not be limited to, sample processing errors that lead to inaccurate bioanalytical results, and/or missed or inaccurate dosing on the day of PK measurements. In the case of a protocol deviation or event, affected PK data may be excluded from any summaries or analyses based on the judgement of the analyst and documented accordingly. Other protocol deviations or events which are not expected to markedly impact the PK results and associated data interpretation may be included for PK analyses, if appropriate.

For the computation of descriptive statistics, serum and plasma concentrations that are below the limit of quantitation (BLQ) will be adjusted based on the conditions below,

- The BLQ value(s) before the first measurable concentration will be treated as zero.
- When an individual BLQ falls between two quantifiable values from EOI of each dose to BI of the subsequent dose, the BLQ value will be set to.
- If two or more BLQ values occur in succession between measurable concentrations from EOI of each dose to BI of the subsequent dose, the measurable concentration(s) after the last BLQ will be considered as missing and BLQ values will be set to LLOQ/2.
- All other BLQ values will be set to LLOQ/2.

Concentrations that are missing will be omitted from the calculation of descriptive statistics. Abnormal concentration results (e.g., anomalous pre-dose concentration value >5% of maximum concentration [ $C_{max}$ ] in the profile) will be excluded from the concentration summaries. If any of the descriptive statistic values (median, mean, minimum and maximum) is below the LLOQ, then the corresponding descriptive statistic values will be presented as BLQ, and standard deviation and CV% values will be presented as not determined at the corresponding timepoint.

Figures of arithmetic mean ( $\pm$  SD) concentration-time profiles will be presented by dose level combining Part 1 and Part 2 participants and by disease indication on linear and semi-logarithmic scales.

Population PK modelling is not in scope of this SAP and will be provided in a separate analysis plan.

#### **4.7.2 Pharmacodynamic Analyses**

Not applicable.

#### **4.7.3 Pharmacogenomic Analyses**

Not applicable.

#### **4.7.4 Immunogenicity Analyses**

The immunogenicity objective is to characterize the immunogenicity of BNT324 and BNT327 based on the immunogenicity analysis set.

The corresponding immunogenicity endpoints include:

- The prevalence and incidence of ADAs against BNT324.
- The prevalence and incidence of ADAs against BNT327.

The categorization of ADA is presented in [Table 7](#) below.

**Table 7: Definitions of ADA categories**

| ADA categories           | Definition                                                                                                                                                                                                                                                    |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADA Positive Sample      | A sample confirmed to be positive for the presence of reactive antibodies. A sample must be positive in both the screening assay and the confirmation assay.                                                                                                  |
| ADA Negative Sample      | A sample that is reported as negative by the screening assay or by the confirmation assay.                                                                                                                                                                    |
| Treatment-emergent ADA   | A sample that meets the definitions of treatment-induced ADA or treatment boosted ADA.                                                                                                                                                                        |
| Treatment-induced ADA    | A sample that is reported as positive after trial treatment administration in a participant with negative or missing baseline values.                                                                                                                         |
| Treatment-boosted ADA    | A sample that increases to a higher level than a pre-defined threshold relative to baseline in a participant with an ADA positive sample before study treatment administration. For this study, a 4-fold increase is the pre-defined threshold.               |
| Treatment-unaffected ADA | A sample that does not increase to a level equal to or below a pre-defined threshold relative to baseline in a participant with an ADA positive sample before study treatment administration. For this study, a 4-fold increase is the pre-defined threshold. |

ADA prevalence, which is the percentage of trial participants who are ADA positive at any time point (baseline or post-baseline), will be summarized. The ADA incidence will also be reported, which is the proportion of trial participants having treatment-emergent ADA at any time point during the trial period.

The number and percentage of participants with ADA positive, negative, or missing samples at baseline and post-baseline will be provided by dose level combining Part 1 and Part 2 participants and by disease indication. The number and percentage of participants who have treatment-induced ADA, treatment-boosted ADA, and treatment-unaffected will also be summarized.

The effect of ADA on PK may be evaluated, if data is available, by summarizing the PK profile of ADA positive trial participants and ADA negative trial participants.

The effect of ADA on efficacy and safety, if data available, will be evaluated by summarizing the key efficacy and safety endpoints in the ADA positive trial participants and ADA negative trial participants.

The assessment of neutralizing ADA prevalence will be evaluated if data is available.

A by-participant listing of ADA results will be produced for the IAS.

#### 4.7.5 Biomarker Analyses

The biomarker objective is to evaluate B7-H3 and PD-L1 expression in tumor tissue and its relationship with efficacy.

The corresponding biomarker endpoints include:

- Expression level of B7-H3 in tumor tissue and its correlation with efficacy.

*Expression level of PD-L1 in tumor tissue and its correlation with efficacy.*

B7-H3 and PD-L1 expression level will be determined using immunohistochemistry (IHC).

Descriptive statistics will be provided for the baseline B7-H3 and PD-L1 using central lab results at selected dose level(s) combining Part 1 and Part 2 participants and by disease indication for the FAS population. A summary of select baseline characteristics by B7H3 and PD-L1 expression level for the FAS will also be provided.

The percentage of tumor cells expressing B7H3 (e.g. membranous staining and cytoplasmic staining), and/or the percentage of tumor-associated immune cells expressing B7H3 will be determined by the central lab.

Association between baseline B7-H3 expressing level and efficacy endpoints (e.g., BOR, ORR, PFS, OS) will be explored using the central lab data as data allows. Analyses will be performed at selected dose level(s) combining Part 1 and Part 2 participants and by disease indications for:

1. Box plots of baseline B7H3 expression level based on response status (confirmed CR/PR vs. others)
2. KM plot for OS, PFS by baseline B7H3 tumor expression level (%1+/2+/3+ cutoffs: <1% vs. 1-49% vs. ≥50%)
3. Waterfall plot of best percentage change from baseline on target lesion sum of diameters by baseline B7H3 tumor expression level (%1+/2+/3+ cutoffs: <1% vs. 1-49% vs. ≥50%)

Analysis may also be performed by pooling selected disease indications.

The percentage of tumor cells expressing PD-L1, and the percentage of tumor-associated immune cells expressing PD-L1 will be determined.

Association between PD-L1 expression level and efficacy endpoints (e.g., ORR, PFS, OS) will be explored as data allows. Analyses will be performed at selected dose level(s) combining Part 1 and Part 2 participants and by disease indications for:

1. Box plots of the percentage of tumor cells expressing PD-L1 based on response status (confirmed CR/PR vs. others)
2. KM plot for OS, PFS by the percentage of tumor cells expressing PD-L1 TPS cutoffs: <1% vs. 1 - 49% vs. ≥50%.
3. Waterfall plot of best percentage change from baseline on target lesion sum of diameters by baseline PD-L1 tumor expression level (TPS cutoffs: <1% vs. 1-49% vs. ≥50%)

Additional biomarker analysis may also be performed as appropriate.

Unless otherwise specified, missing biomarker data will be treated as such, and no imputed values will be derived.

By-participant listings may be also produced.

#### **4.7.6 Health Outcomes Analyses – Part 2 Cohort 1 and 2**

The objective is to evaluate HRQoL endpoints using the EORTC QLQ-C30, EORTC QLQ-LC29, PGIS, and PGIC in participants enrolled in Part 2 Cohort 1 and 2.

The corresponding endpoints include:

- Change from baseline in EORTC QLQ-C30 Global Health status/QoL scale, functional scales, and symptom scales/item.
- Change from baseline in EORTC QLQ-LC29 symptom scales/item.
- Change from baseline in PGIS overall status.
- PGIC post-baseline overall status from C2D1.

The analysis of health outcome endpoints will be based on the FAS. Missing health outcome data will be treated as such, and no imputed values will be derived.

### **Completion rate**

The completion rate of each instrument will be summarized by visit and by cohort and treatment arm. The numbers and percentages of completed and missing data for each visit will be summarized.

The completion rate at a specific visit for a given instrument is defined as the number of treated participants who complete at least one item on that PRO instrument over the number of treated participants in the FAS.

#### **4.7.6.1 EORTC QLQ-C30**

The EORTC QLQ-C30 was developed by the EORTC Quality of Life Group to assess HRQoL, functioning, and symptoms in cancer clinical trials. It is a 30-item self-administered questionnaire for all cancer types. Questions are grouped into five multi-item functional scales (physical, role, emotional, cognitive, and social), three multi-item symptom scales (fatigue, pain, and nausea/vomiting), a 2-item global QoL scale, five single-items assessing additional symptoms commonly reported by cancer participants (dyspnea, loss of appetite, insomnia, constipation, and diarrhea), and one item on the financial impact of the disease (. All but two questions have 4-point scales: "Not at All", "A Little", "Quite a Bit", and "Very Much". The two questions concerning global health status and QoL have 7-point scales with responses ranging from "Very poor" to "Excellent". For each of the 15 domains, final scores are transformed such that they range from 0 to 100, where higher scores indicate better functioning, better HRQoL, or greater level of symptoms.

Both multi-item scales and single-item measures are shown in [Table 8](#).

**Table 8: Scoring the EORTC QLQ-C30**

| Scale                    | Item Range | Raw Score                    |
|--------------------------|------------|------------------------------|
| Global health status/QoL | 6          | Mean of (Q29, Q30)           |
| Functional Scales        |            |                              |
| Physical Functioning     | 3          | Mean of (Q1, Q2, Q3, Q4, Q5) |
| Role Functioning         | 3          | Mean of (Q6, Q7)             |
| Emotional Functioning    | 3          | Mean of (Q21, Q22, Q23, Q24) |
| Cognitive Functioning    | 3          | Mean of (Q20, Q25)           |
| Social Functioning       | 3          | Mean of (Q26, Q27)           |
| Symptom Scales/Items     |            |                              |
| Fatigue                  | 3          | Mean of (Q10, Q12, Q18)      |
| Nausea and Vomiting      | 3          | Mean of (Q14, Q15)           |



| Scale                  | Item Range | Raw Score         |
|------------------------|------------|-------------------|
| Pain                   | 3          | Mean of (Q9, Q19) |
| Dyspnea                | 3          | Q8                |
| Insomnia               | 3          | Q11               |
| Appetite Loss          | 3          | Q13               |
| Constipation           | 3          | Q16               |
| Diarrhea               | 3          | Q17               |
| Financial Difficulties | 3          | Q28               |

Once the raw scores are calculated, a linear transformation to 0 to 100 is applied to obtain the particular score as follows:

- For global health status/QoL and symptom scales/items:  $\text{Score} = \{(\text{Raw score} - 1) / \text{Range}\} * 100$
- For functional scales:  $\text{Score} = \{1 - (\text{Raw score} - 1) / \text{Range}\} * 100$

Range is the difference between the maximum possible value of raw score and the minimum possible value. The QLQ-C30 has been designed so that all items in any scale take the same range of values. Therefore, the range of raw score equals the range of the item values. Most items are scored 1 to 4, giving range = 3. The exceptions are the items contributing to the global health status / QoL, which are 7-point questions with range = 6.

After the linear transformation, the score of each scale/item has a range of 0-100. A high score represents a high level of functioning for QoL and a functional scale, and a high level of symptomatology/problem for a symptom scale/items. If less than half of the constituent items have been answered for a multi-item subscale, that subscale will be considered as missing. Single item subscales will be considered missing if the constituent item is incomplete.

Change from baseline will be summarized as descriptive statistics by cohort, treatment arm, and visit as appropriate.

#### 4.7.6.2 EORTC QLQ-LC29

The QLQ-LC29 is a supplementary questionnaire module to be employed in conjunction with the QLQ-C30. Questions are grouped into five multi-item scales (coughing, shortness of breath, hair problems, fear of progression, and surgery-related symptoms) and fifteen single items (coughed up blood, sore mouth or tongue, problems swallowing, tingling hands or feet, chest pain, pain in arm or shoulder, pain in other parts of body, allergic reactions, burning or sore eyes, dizziness, splitting fingernails or toenails, skin problems, problems speaking, decrease in physical capabilities and Weight loss problem). All multi-item scales and single items have 4-point scales: "Not at All", "A Little", "Quite a Bit", and "Very Much". final scores are transformed such that they range from 0 to 100, where higher scores indicate better functioning, better HRQoL, or greater level of symptoms.

Both multi-item scales and single-item measures are shown in [Table 9](#).



**Table 9: Scoring the EORTC QLQ-LC29**

| Multi-item Scales                 | Item Range | Raw Score                         |
|-----------------------------------|------------|-----------------------------------|
| Coughing                          | 3          | Mean of (Q31, Q52)                |
| Shortness of Breath               | 3          | Mean of (Q33, Q34, Q35)           |
| Hair Problems                     | 3          | Mean of (Q39, Q50)                |
| Fear of Progression               | 3          | Mean of (Q49, Q51)                |
| Surgery-related Symptoms          | 3          | Mean of (Q55, Q56, Q57, Q58, Q59) |
| Single Items                      |            |                                   |
| Coughed up Blood                  | 3          | Q32                               |
| Sore mouth or Tongue              | 3          | Q36                               |
| Problems Swallowing               | 3          | Q37                               |
| Tingling Hands or Feet            | 3          | Q38                               |
| Chest Pain                        | 3          | Q40                               |
| Pain in Arm or Shoulder           | 3          | Q41                               |
| Pain in Other Parts of Body       | 3          | Q42                               |
| Allergic Reactions                | 3          | Q43                               |
| Burning or Sore Eyes              | 3          | Q44                               |
| Dizziness                         | 3          | Q45                               |
| Splitting Fingernails or Toenails | 3          | Q46                               |
| Skin Problems                     | 3          | Q47                               |
| Problems Speaking                 | 3          | Q48                               |
| Decrease in Physical Capabilities | 3          | Q53                               |
| Weight Loss Problem               | 3          | Q54                               |

Once the raw scores are calculated, a linear transformation to 0 to 100 is applied to obtain the particular score as follows:

- Symptom scales:  $\text{Score} = \{(\text{Raw score}-1)/\text{Range}\} \times 100$

Range is the difference between the maximum possible value of row score and the minimum possible value. The QLQ-LC29 has been designed so that all items in any scale take the same range of values. Therefore, the range of raw score equals the range of the item values. All items are scored 1 to 4, giving range = 3.

After the liner transformation, the score of each scale/item has a range of 0-100. A high score for the symptom scales and single items represents a high level of symptomatology or problems. If less than half of the constituent items have been answered for a multi-item subscale, that subscale will be considered as missing. Single item subscales will be considered missing if the constituent item is incomplete.

Change from baseline will be summarized as descriptive statistics by cohort, treatment arm, and visit as appropriate.

#### 4.7.6.3 PGIS

PGIS is a questionnaire to assess the participant's overall status since starting to receive the study treatment. The overall status has 5 scales: "None", "Mild", "Moderate", "Severe",

and “Very Severe”. A shift table will be presented to summarize overall status for participants at baseline and the worst post-baseline overall status by cohort and treatment arm.

#### 4.7.6.4 PGIC

PGIC is a questionnaire to assess the participant’s overall status since C2D1. The overall status has 7-point scale: [1] “Very Much Improved”, [2] “Much Improved”, [3] “Minimally Improved”, [4] “No Change”, [5] “Minimally Worse”, [6] “Much Worse”, and [7] “Very Much Worse”, where lower score indicates better overall status. A shift table will be presented to summarize post-baseline overall status by cohort, treatment arm, and visit.

### 4.8 Safety Analyses

Safety data that will be summarized includes drug exposure, adverse events, clinical laboratory assessments, vital signs LVEF, and ECGs. All safety analyses will be based on the FAS and will be summarized by trial part, dose level, cohort and treatment arm as appropriate. Participants will be assigned to the trial arm according to the actual treatment they have received (“as treated”).

There will be no formal statistical comparisons between the trial arms for safety data.

#### 4.8.1 Extent of Exposure

Administration of IMP will be presented as recorded on the “Study Drug Administration – BNT324” and “Study Drug Administration – BNT327” eCRFs.

The number and percentage of participants not treated and treated with each dose and combined will be summarized by trial part, dose level, cohort, and treatment arm as appropriate.

The following dose exposure variables will be derived and summarized descriptively for each trial part, dose level, cohort, and treatment arm as appropriate, both separately and combined for BNT324 and BNT327, respectively based on the FAS:

- Number of cycles,
- treatment duration (weeks),
- cumulative dose,
- actual dose intensity (mg/kg/week),
- relative dose intensity (%),
- Category of relative dose intensity (<60%, ≥60 to <80%, ≥80 to <90%, ≥90 to <110%, ≥110%),

where the above variables are derived as follows:

- Treatment duration (weeks) is defined as: (Date of last administration – Date of first administration + 21)/7.
- Cumulative Dose (mg) is defined as sum of all administered doses.
- Dose Intensity (mg/kg/week) is defined as Dose level (mg/kg)/Treatment Duration (weeks), where actual dose intensity is defined as the sum of actual dose levels (mg/kg)/Treatment Duration (weeks) and planned dose intensity is defined as the planned dose level (mg/kg)/3.

- Relative Dose Intensity (RDI) is defined as  $100 \times (\text{Actual/Planned Dose Intensity})$  and expressed as a percentage. The relative dose intensity will be presented categorically (i.e., number and percentage of participants with relative dose intensity of <60%, ≥60 to <80%, ≥80 to <90%, ≥90 to <110%, ≥110%). Moreover, dose modifications may be summarized as appropriate.

#### 4.8.2 Adverse Events

AEs are those recorded by the investigator on the eCRF, and will be analyzed by trial part, dose level, cohort, and treatment arm as appropriate based on the FAS.

##### 4.8.2.1 DLT

The definition of DLT and analytical approach is provided in Section 4.4.1.

##### 4.8.2.2 Treatment-Emergent Adverse Events

The definition of TEAE and corresponding analytical approach are provided in the Section 4.4.1.

In addition, individual summaries (number of participants and percentages) of AEs will be presented as specified in sections below.

##### TEAEs

- TEAEs by system organ class and preferred term.
- TEAEs by system organ class, preferred term, and maximum toxicity grade.
- Grade 3 to 5 TEAEs by system organ class and preferred term.
- Related TEAEs by system organ class and preferred term.
- Related TEAEs by system organ class and preferred term, and maximum toxicity grade.
- Related Grade 3 to 5 TEAEs by system organ class and preferred term.

##### Serious TEAEs

- Serious TEAEs by system organ class and preferred term.
- Serious Related TEAEs by system organ class and preferred term.

##### TEAEs leading to dose modification, reduction, interruption, discontinuation.

##### *Dose reduction*

- TEAEs leading to dose reduction (only for BNT324) by system organ class and preferred term.
- Related TEAEs leading to dose reduction (only for BNT324) by system organ class and preferred term.

##### *Drug interruption*

- TEAEs leading to drug interruption by system organ class and preferred term.
- Related TEAEs leading to drug interruption by system organ class and preferred term.

### *Drug discontinuation*

- TEAEs leading to drug discontinuation by system organ class and preferred term.
- Related TEAEs leading to drug discontinuation by system organ class and preferred term.

### TEAEs resulting in death

- TEAEs resulting in death by system organ class and preferred term.
- Related TEAEs resulting in death by system organ class and preferred term.

Additional TEAE summary tables at specific incidence rate cutoffs by PT by decreasing incidence may be provided as appropriate.

Additional TEAE analysis (e.g. TEAEs of Custom MedDRA Queries or special categories, exposure adjusted TEAEs, time to AE onset) may also be performed as appropriate.

Non-serious TEAEs occurring in at least 5% of participants in any trial arm will be summarized by SOC and preferred term. The number of participants and number of events will be presented for this summary.

See [Appendix 7](#) for handling of partial dates for AEs. In cases where it is not possible to define an AE as treatment-emergent or not, the AE will be classified by the worst case, i.e., treatment-emergent.

Uncoded events and procedures at the time of analysis will be assigned the string "UNCODED" as the SOC, and the reported term will be used as the preferred term so they can be included in the summary tables. In the final dataset, all the AEs and procedures should be coded. Uncoded medications at the time of analysis will be assigned the string "UNCODED" as the Anatomical Therapeutic Chemical coding dictionary (ATC) summary level(s) and the verbatim term will be used as the preferred name, so they can be included in the summary tables. In the final dataset, all the medications will be coded.

When summarizing events by causality and severity by participants, each participant will be counted only once within a system organ class and within a preferred term by using the event with the greatest relationship and highest severity within each classification. If relatedness to trial treatment is missing for a TEAE it will be imputed as the worst case and assumed to be related. TEAEs with a missing toxicity grade will be analyzed as Grade 3.

By-participant listings of AEs will also be created as appropriate.

### **Summary of Deaths**

A summary of on-treatment (on or after date of first dose of trial treatment administration until 90 days post last dose) deaths and primary reason for death and additional summary of all deaths (on-treatment + during follow-up) will be provided along with primary reason for death by trial arm. Time from last dose of trial treatment to death will also be summarized. Death will be summarized in the FAS.

All deaths including those that occurred on-treatment and during follow-up will be listed, post-treatment deaths will be flagged.

#### 4.8.2.3 Adverse Events of Special Interest

A treatment-emergent AE of special interest (AESI), serious or non-serious, is one of scientific and medical concerns specific to the sponsor's product or program, for which ongoing monitoring and rapid communication by the investigator to the sponsor are appropriate. For the list of AESIs for this trial refer to CTP v3.1 Section 8.5.8.

AESIs will be identified as investigator assessment.

An overall treatment emergent AESI summary will be provided including a summary for each trial part by cohort/treatment group of the number and percentage of participants with at least one TEAE of:

- AESI.
- Drug-related AESI.
- Procedure-related AESI.
- Serious AESI.
- Serious drug-related AESI.
- Grade  $\geq 3$  AESI.
- Drug -related Grade  $\geq 3$  AESI.
- AESI leading to dose reduction.
- AESI leading to drug interruption.
- AESI leading to drug delay
- AESI leading to drug discontinuation.
- Drug-related AESI leading to:
  - dose reduction.
  - drug interruption.
  - drug delay.
  - drug discontinuation.
- Death due to an AESI.
- Death due to a drug-related AESI.

A summary of AESI and related AESI by AESI category, PT, and maximum toxicity grade will be presented.

A summary of adjudicated ILD/pneumonitis AESIs will be provided. For ILD/pneumonitis AESIs, maximum toxicity grade, time to onset, drug relatedness, and outcome will be summarized for both by investigator and adjudicated events.

Additional AESI analyses may also be performed as appropriate.

Drug-related AESIs will be split (where applicable) by BNT324 only, BNT327 only, BNT324 or BNT327 (if AESI was related to either drug), BNT324 and BNT327 (if AESI was related to both drugs). AESIs leading to dose reduction, drug interruption, drug delay, and drug discontinuation will be presented in the same manner. For drug-related AESI leading to dose modifications (i.e., dose reduction, drug interruption, drug delay and drug discontinuation), the AESIs that are considered either related to BNT324 or BNT327 will be included.

By-participant listings will also be created.

### 4.8.3 Laboratory Data

Clinical laboratory data to be summarized based on the FAS, including hematology, blood biochemistry tests, coagulation tests, and quantitative urinalysis tests, will be analyzed for each trial part, dose level, cohort, and treatment arm if applicable, using data reported on the following eCRFs: "Hematology – Local", "Chemistry – Local", "Coagulation – Local", "Urinalysis – Local", and "Pregnancy Test". Refer to BNT324-01 CTP v3.1, Section 8.4.7 for the complete list of clinical laboratory parameters.

For any laboratory value reported using a non-numeric qualifier (e.g., less than (<xx.x) a certain value, or greater than (>yy.y) a certain value), the inequality sign will be removed and the resulting numeric value will be imputed as the observed value, or for change from baseline calculations. When evaluating the value to determine the CTCAE severity grade, the numeric values will be used; however, if the numeric value is equal to a cut-off value, the inequality signs (<xx.x or >yy.y) will be considered.

Laboratory test grade based on the NCI-CTCAE version 5.0 will be derived for all parameters with NCI-CTCAE criteria. Shift tables will be presented to summarize laboratory test results at baseline and worst post-baseline toxicity grade. A summary of laboratory test results worst post-baseline toxicity grade for participants who worsened post-baseline will also be presented (see [Appendix 5](#) for reference to the criteria).

Shift tables will be presented to summarize laboratory test results at baseline and worst post-baseline value in clinical significance category by trial part, dose level, cohort, treatment arm, if available. Normal ranges established by the local laboratory will be used to determine whether the value is lower (L), within normal (N) or higher (H) than the reference range for that parameter.

Clinical laboratory values for each parameter will be plotted over time.

A listing of abnormal laboratory tests and quantitative urinalysis results at any evaluation may also be provided. Determination of clinical significance made by the investigator will be included in the listing, if available. In addition, any windowed value used for by-visit analysis will be flagged in the listing.

A listing of qualitative urinalysis results at any evaluation may also be provided.

#### Liver Function Tests - combined elevations of aminotransferases and bilirubin

Participants who present with specific laboratory abnormalities should be evaluated further as potential drug-induced liver injury (Hy's law) cases to definitively determine the etiology of the abnormal laboratory values.

A summary of number and percentage of participants will be produced for the following:

- AST  $\geq 3 \times$  ULN.
- ALT  $\geq 3 \times$  ULN.
- AST or ALT  $\geq 3 \times$  ULN.
- Total bilirubin  $\geq 2 \times$  ULN.
- ALP  $\geq 2 \times$  ULN.
- AST or ALT  $\geq 3 \times$  ULN and total bilirubin  $\geq 2 \times$  ULN.
- AST or ALT  $\geq 3 \times$  ULN and total bilirubin  $\geq 2 \times$  ULN and ALP  $< 2 \times$  ULN.

For overall summary and summaries combining multiple tests, the tests (AST, ALT, bilirubin, and ALP) should be based on values obtained at the same visit.

Additional liver function test categories may also be summarized as appropriate.

A scatter plot will be produced of maximum post-baseline total bilirubin ( $\times$  ULN) against the maximum post-baseline ALT or AST ( $\times$  ULN) to help identify potential cases of drug-induced liver injury or Hy's law.

A listing of liver function tests results of participants with abnormal liver function test results meeting the above criteria (all above bullets with a " $\geq$ " sign) will also be provided.

### **Other laboratory data**

Other laboratory data to be assessed includes thyroid function test, myocardial enzyme spectrum examination, and serum virology. Results will be summarized with descriptive statistics by visit by trial part, dose level, cohort, and treatment arm as appropriate based on the FAS. The data may be listed for all participants by trial part, dose level, cohort, and treatment arm as appropriate, using information available on "Chemistry – Local", "Serology – Local" eCRFs.

#### **4.8.4 Vital Signs**

Vital sign parameters to be summarized based on the FAS include systolic blood pressure (mmHg), diastolic blood pressure (mmHg), respiratory rate (breath/min), body temperature ( $^{\circ}$ C), pulse rate (bpm), and oxygen saturation (mmHg) will be assessed at the timepoints as indicated in the SoA in [Appendix 1](#), by trial part, dose level, cohort, and treatment arm as appropriate.

Vital signs will be summarized (as indicated on the "Vital Signs" eCRF) for actual value for each visit as well as the change from baseline for each post-baseline visit using descriptive statistics.

A by-participant listing may also be produced for participants who reported abnormal findings, including clinical significance.

#### **4.8.5 Other Safety Assessments**

##### **4.8.5.1 12-lead Electrocardiogram**

ECG parameters to be summarized based on the FAS include QT, QTcF, PR, and QRS intervals in msec and heart rate (beats per minute [bpm]). ECG parameters will be summarized descriptively by visit for actual values and changes from baseline by trial part, dose level, cohort, and treatment arm as appropriate.

A 12-lead ECG assessment will be conducted at screening, at the first day of each cycle prior to trial treatment administration, at EoT Visits, and at Safety follow-up (FU).

At screening, ECGs will be obtained locally in triplicate (three individual ECG tracings should be obtained in succession within 10 min). Subsequent ECGs (at scheduled visits) will be obtained locally in triplicate only if abnormalities are noted (i.e. if first reading is abnormal, two more readings will be taken). Additional assessments may be performed when clinically indicated.



Electrocardiogram data will be obtained from local reading as recorded in the “Electrocardiogram - Local – Log (Triplicate)” and “Electrocardiogram - Local – Log” eCRFs:

Note: at the screening visit, local ECGs will be obtained as a triplicate. At all post screening unscheduled assessments, local ECGs will be obtained as a single reading or as a triplicate.

ECG parameters to be reported are ventricular heart rate (beats/minute), ventricular rate (beats/minute), PR interval (msec), QRS duration (msec), QT interval (msec), and QT interval corrected for heart rate by Fridericia's formula (QTcF) (msec).

$$QTcF = QT (RR)^{-(1/3)}$$

ECGs will be interpreted by the reader as “Normal”, “Abnormal”, “Intermediate”, and “Not evaluable” For abnormal ECGs, further assessment will be conducted to categorize them as clinically significant or not. The number and percentage of participants in each ECG interpretation category will be summarized by trial part, dose level, cohort, and treatment arm as appropriate.

For the purpose of analysis, local ECGs (single and triplicate) will be combined. The average of the triplicate results of triplicate central ECG at each timepoint will be first calculated and then pooled with single ECG results for the analysis.

Local ECGs will be summarized as raw values and change from baseline values at each analysis visit and will be provided for HR (beats/min), RR interval (msec), PR Interval (msec), QRS Interval (msec), QT Interval (msec), and QTcF Interval (msec).

The number and percentage of participants in the following QTcF Interval categories will be summarized:

- QTcF value at any time post-baseline (>450 msec, >470 msec, >480 msec, >500 msec).
- QTcF increase at any time post-baseline (>30 msec, >60 msec, >90 msec).
- QTcF value and QTcF increase at any time post-baseline (Value >450 msec and increase >30 msec, Value >470 msec and increase >30 msec, Value >500 msec and increase >60 msec).

Assessment of local ECG results (single and triplicate) may be presented in a by-participant data listing for the FAS as appropriate.

Additional ECG analysis may also be provided as appropriate.

#### **4.8.5.2 Physical Examination**

No statistical summaries will be provided for physical examinations. Physical examination symptomatic-directed physical exam data may be presented in by-participant data listings as indicated on the “Physical Examination - Traditional” and “Physical Examination” eCRFs for the FAS.

#### **4.8.5.3 Pregnancy Test**

Pregnancy data are collected on the “Pregnancy Test” eCRF. Pregnancy data, including urine and serum pregnancy tests results (Positive/Negative/Indeterminate) will be

presented in by-participant data listings for all women of childbearing potential (identified through the 'Childbearing Potential Status' eCRF) for the FAS.

#### **4.8.5.4 ECHO or MUGA Scan**

The following summaries will be provided for LVEF data using results indicated on the "ECHO or MUGA Scan" eCRF for the FAS:

- Actual and change from baseline by visit.
- Shift from baseline according to worst post-baseline result (i.e., the least absolute value) in each clinical significance category (Normal; Abnormal, NCS; Abnormal, CS) for overall interpretation of LVEF measurements (using data from on-treatment period only).
- The number and percentage of participants in the following LVEF categories will be summarized:
  - LVEF decrease of  $\geq 10$  percentage points from baseline to an absolute value  $\geq 45\%$  and  $< 50\%$
  - LVEF decrease of  $\geq 10$  and  $\leq 15$  percentage points from baseline to an absolute value of  $\geq 40\%$  and  $< 45\%$
  - LVEF decrease of  $\geq 15$  percentage points from baseline to an absolute value of  $\geq 50\%$
  - LVEF decrease of  $\geq 16$  and  $\leq 20$  percentage points from baseline to an absolute value of  $\geq 40\%$  and  $< 45\%$
  - LVEF decrease of  $\geq 20$  percentage points from baseline or to an absolute value of  $< 40\%$

Additional LVEF categories may also be summarized as appropriate.

ECHO and MUGA scan data may also be presented in by-participant data listing as appropriate.

#### **4.8.5.5 Ophthalmologic examination**

By-participant data listings reporting ophthalmic exam results as indicated on the "Ophthalmic Exam" eCRF will be provided.

#### **4.8.5.6 ECOG**

ECOG as indicated by the investigator on the "ECOG" eCRF will be summarized by performance status for each visit by trial part/cohort/treatment group as categorized below for the FAS:

- Score 0: Fully active, able to carry on all pre-disease performance without restriction.
- Score 1: Restricted in physically strenuous but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work).
- Score 2: Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.
- Score 3: Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.

- Score 4: Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
- Score 5: Dead.

A shift table will be produced showing the number and percentage of participants shifting from baseline to a worst post-baseline result (0, 1, 2, 3, 4, 5) in each category (using data from on-treatment period only).

#### 4.9 Subgroup Analyses of Interest

ORR may be provided based on the FAS, by disease indications, if more than 40 participants were enrolled at selected dose level(s) for the subgroup factors (based on actual baseline values from EDC or central lab data) include:

- B7-H3 expression: %1+/2+/3+ ≤1%, 1-49%, and ≥50%.
- PD-L1 expression: TPS ≤1%, 1-49%, and ≥50%.
- Number of prior lines of therapy: 1, 2+.
- Number of prior lines of therapy (1, 2, 3+) in advanced or metastatic settings,
- Number of prior lines of therapy (1, 2+) in adjuvant or neo-adjuvant settings,
- Age group: <65, ≥65 years.
- Race: White, Asian, Black or African American, Other.  
Note: 'Other' includes American Indian or Alaska Native, Native Hawaiian or Other Pacific Islanders, and/or Other as indicated on 'Demography' eCRF.
- Region: Asia-Pacific, North America, Europe, Rest of World.
- ECOG: 0, 1, 2.
- Smoking status: never, current, former

Additional subgroup analyses may be conducted as appropriate.

## 5 INTERIM ANALYSIS AND ANALYSIS SEQUENCE

No formal interim statistical analysis is planned. However, the safety and efficacy data will be monitored on an ongoing basis.

For Part 1, safety data will be continuously monitored for each cohort. Additional participants may be backfilled to selected dose levels. The dose levels moving to Part 2 will be determined based on the totality of the data.

Part 2 will further evaluate the anti-tumor activity and safety profile of the IMPs. Based on the totality of the available data (e.g., safety, efficacy, PK etc.), the optimal dose level will be selected for Cohort 1 and Cohort 2 target indication, and additional 20 participants will be enrolled in the selected optimal dose arm for each of Cohort 1 and 2.

For each of the Cohorts 3-7 in Part 2, an early futility check on ORR results will be performed after the first 10 participants become response evaluable in the corresponding cohort. A cohort may be considered futile and stopped for further enrollment if it does not meet the criteria as listed in [Table 3](#). Enrollment will not pause when conducting the analysis.

In addition, an expansion evaluation analysis is planned for each of the Cohorts 3-7 after the first 20 participants become response evaluable in each of the cohorts. To facilitate the expansion decision-making, the observed ORR (unconfirmed) results from each cohort will

be compared to the lower and higher target ORR for each cohort. A cohort will be stopped for further enrolment if the posterior probability of ORR (unconfirmed) greater than the higher target value is lower than or equal to 5%. A cohort will be considered promising for further enrolment if the posterior probability of ORR (unconfirmed) greater than the lower and higher target values are greater than 80% and 5%, respectively. The lower and higher target ORR values and the corresponding decision criteria for each cohort are listed in [Table 4](#). A non-informative Beta (0.001, 0.001) prior will be applied. In addition, depending on the totality of the data, selected cohorts may be chosen to enroll up to 40 participants per cohort. Of note, these criteria are non-binding and the totality of the available data (e.g., safety, efficacy, PK, etc.) will be reviewed by the sponsor to determine expansion or not. This potential expansion may utilize dosing at the RP2D or RP2D-1. Enrolment for other ongoing cohorts will not pause when conducting the analysis in each cohort.

**Table 10: Probability of not Expanding based on assumed true ORR**

| Cohort | Indication                             | Lower Target ORR | Higher Target ORR | Not Expansion Criterion N=20 | True ORR | Probability of Not Expanding N=20 |
|--------|----------------------------------------|------------------|-------------------|------------------------------|----------|-----------------------------------|
| 3      | NSQ NSCLC, AGA-, 2L, post-chemo +/- IO | 20%              | 35%               | $\leq 15\%$ (3/20)           | 30%      | 11%                               |
|        |                                        |                  |                   |                              | 35%      | 5%                                |
| 4      | SQ, AGA-, 1L                           | 50%              | 60%               | $\leq 40\%$ (8/20)           | 55%      | 13%                               |
|        |                                        |                  |                   |                              | 60%      | 6%                                |
| 5      | SQ, AGA-, 2L, post-chemo +/- IO        | 20%              | 30%               | $\leq 15\%$ (3/20)           | 25%      | 23%                               |
|        |                                        |                  |                   |                              | 30%      | 11%                               |
| 6      | NSQ, AGA+, post-TKI                    | 40%              | 50%               | $\leq 30\%$ (6/20)           | 45%      | 13%                               |
|        |                                        |                  |                   |                              | 50%      | 6%                                |
| 7      | ES-SCLC, 1L                            | 70%              | 80%               | $\leq 60\%$ (12/20)          | 75%      | 10%                               |
|        |                                        |                  |                   |                              | 80%      | 3%                                |

The primary analysis of the trial will be performed when all participants have either discontinued the trial or have been (or have had the chance to be) followed for at least 18 months from the start of IMP, whichever occurs first.

## 6 SAMPLE SIZE DETERMINATION

Part 1 is the dose escalation part. The sample size for trial Part 1 is driven by the BOIN design and will include up to 9 DLT-evaluable participants per dose level depending on the occurrence of DLTs, with a maximum of 54 DLT-evaluable participants in total. Non-DLT-evaluable trial participants will be replaced. Per sponsor's decision, among the dose levels that have been demonstrated to be safe and well tolerated, or at their lower dose levels, additional participants may be backfilled to selected dose levels to reach a total of up to 80 participants in each line (first line or second line) of NSCLC disease indication and a total of up to 60 participants in the second line SCLC disease indication (if they meet the criteria for Part 2). These participants will not be counted towards the total sample size of the BOIN design. A total of up to 274 participants will be enrolled in Part 1, including backfill participants.

Part 2 will further evaluate the anti-tumor activity and safety profile of the IMPs. The total sample size for Part 2 is up to 320. In the randomization DO Cohort 1 and Cohort 2, ~40 participants will be enrolled in each cohort and randomized at a 1:1 ratio to one of the two dose levels selected from Part 1. The totality of the available data (e.g., safety, efficacy, PK etc.) will be reviewed to select the optimal dose. After the optimal dose is selected, an additional 20 participants in each cohort may be enrolled in the selected optimal dose to further assess the efficacy and safety profile. For the other disease Cohorts 3-7, up to 40 participants are planned to be enrolled in each cohort. The 90% exact CIs for ORR at N = 20 and 40 at various observed ORR are provided in Table 11. In addition, with 20 participants per dose arm or 40 participants per dose arm or cohort, if the true incidence rate is 15% for a specific AE, then there is a 96.1% and 99.8% probability of observing at least one AE, respectively.

It is thus anticipated that the trial will enroll up to 594 participants, including backfill participants in Part 1.

**Table 11: Precision estimate of ORR**

| N  | ORR | 90% exact CI lower limit | 90% exact CI upper limit |
|----|-----|--------------------------|--------------------------|
| 20 | 30% | 14.0%                    | 50.8%                    |
| 20 | 40% | 21.7%                    | 60.6%                    |
| 20 | 50% | 30.2%                    | 69.8%                    |
| 20 | 60% | 39.4%                    | 78.3%                    |
| 20 | 70% | 49.2%                    | 86.0%                    |
| 20 | 80% | 59.9%                    | 92.9%                    |
| 40 | 30% | 18.3%                    | 44.0%                    |
| 40 | 40% | 26.9%                    | 54.2%                    |
| 40 | 50% | 36.1%                    | 63.9%                    |
| 40 | 60% | 45.8%                    | 73.1%                    |
| 40 | 70% | 56.0%                    | 81.7%                    |
| 40 | 80% | 66.8%                    | 89.6%                    |

## 7 CHANGES FROM AND ADDITIONS TO METHODS PLANNED IN THE PROTOCOL

Not applicable.

## 8 REFERENCES

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## 9 SUPPORTING DOCUMENTATION

### Appendix 1 Schedule of Procedures/Assessments

The schedule of assessments for the study is presented below.

**Table 12: Schedule of Activities - screening and treatment period**

|                                                         | Screening period <sup>a</sup> | Treatment period  |     |        |        |         |     |         |     |         |                               |     |
|---------------------------------------------------------|-------------------------------|-------------------|-----|--------|--------|---------|-----|---------|-----|---------|-------------------------------|-----|
|                                                         |                               | Cycle 1 (21 days) |     |        |        | Cycle 2 |     | Cycle 3 |     |         | Cycle 4 and subsequent cycles |     |
|                                                         |                               | D1                |     | D8     | D15    | D1      |     | D1      |     | D8      | D1                            |     |
|                                                         |                               | BI                | EOI |        |        | BI      | EOI | BI      | EOI |         | BI                            | EOI |
| Visit window                                            | D -28 to Day -1               | -                 |     | ±1 day | ±1 day | ±3 days |     | ±3 days |     | ±3 days | ±3 or ±4 days <sup>b</sup>    |     |
| General conditions and safety                           |                               |                   |     |        |        |         |     |         |     |         |                               |     |
| Informed consent                                        | X                             |                   |     |        |        |         |     |         |     |         |                               |     |
| Assessment of inclusion/exclusion criteria <sup>c</sup> | X                             | X                 |     |        |        |         |     |         |     |         |                               |     |
| Demographic data                                        | X                             |                   |     |        |        |         |     |         |     |         |                               |     |
| Tumor history/medical history <sup>d</sup>              | X                             |                   |     |        |        |         |     |         |     |         |                               |     |
| Ophthalmological examination <sup>e</sup>               | X                             |                   |     |        |        |         |     | X       |     |         |                               |     |
| Full physical examination <sup>f</sup>                  | X                             | X                 |     |        |        |         |     |         |     |         |                               |     |
| Symptom-directed physical examination <sup>f</sup>      |                               |                   |     | X      | X      | X       |     | X       |     | X       | X                             |     |
| ECOG PS scoring <sup>g</sup>                            | X                             | X                 |     |        |        | X       |     | X       |     |         | X                             |     |
| Body height                                             | X                             |                   |     |        |        |         |     |         |     |         |                               |     |
| Body weight <sup>h</sup>                                | X                             | X                 |     |        |        | X       |     | X       |     |         | X                             |     |
| Vital signs <sup>i</sup>                                | X                             | X                 | X   | X      | X      | X       | X   | X       | X   | X       | X                             |     |
| SpO <sub>2</sub> <sup>j</sup>                           | X                             | X                 |     |        |        | X       |     | X       |     |         | X                             |     |
| 12-lead ECG <sup>k</sup>                                | X                             | X                 | X   | X      | X      | X       | X   | X       | X   |         | X                             |     |



|                                                                             | Screening period <sup>a</sup> | Treatment period                                |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
|-----------------------------------------------------------------------------|-------------------------------|-------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------|--------|---------|-----|---------|-----|---------|-------------------------------|-----|
|                                                                             |                               | Cycle 1 (21 days)                               |     |                                                                                                                       |        | Cycle 2 |     | Cycle 3 |     |         | Cycle 4 and subsequent cycles |     |
|                                                                             |                               | D1                                              |     | D8                                                                                                                    | D15    | D1      |     | D1      |     | D8      | D1                            |     |
|                                                                             |                               | BI                                              | EOI |                                                                                                                       |        | BI      | EOI | BI      | EOI |         | BI                            | EOI |
| Visit window                                                                | D -28 to Day -1               | -                                               |     | ±1 day                                                                                                                | ±1 day | ±3 days |     | ±3 days |     | ±3 days | ±3 or ±4 days <sup>b</sup>    |     |
| Echocardiography/MUGA <sup>l</sup>                                          | X                             |                                                 |     |                                                                                                                       |        |         |     |         |     |         | X                             |     |
| AE/SAE monitoring <sup>m</sup>                                              | X                             | X                                               |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Prior and concomitant medications <sup>n</sup>                              | X                             | X                                               |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Pregnancy counseling <sup>o</sup>                                           | X                             |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Laboratory tests                                                            |                               |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Hematology and blood biochemistry <sup>p</sup>                              | X                             | X                                               |     | X                                                                                                                     | X      | X       |     | X       |     |         | X                             |     |
| Coagulation, thyroid function test, and myocardial enzyme test <sup>q</sup> | X                             | X                                               |     | X                                                                                                                     | X      | X       |     | X       |     |         | X                             |     |
| Urinalysis <sup>r</sup>                                                     | X                             |                                                 |     | X                                                                                                                     |        | X       |     | X       |     |         | X                             |     |
| Stool routine and occult blood test                                         | X                             | Conduct if deemed necessary by the investigator |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Pregnancy test <sup>s</sup>                                                 | X                             | X                                               |     |                                                                                                                       |        | X       |     | X       |     |         | X                             |     |
| Serum virology test <sup>t</sup>                                            | X                             |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Efficacy evaluation                                                         |                               |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Tumor evaluation <sup>u</sup>                                               | X                             |                                                 |     | Once every 6 weeks (±7 days) for the first 36 weeks after first dose of IMP, and every 12 weeks (±14 days) thereafter |        |         |     |         |     |         |                               |     |
| Randomization/enrollment                                                    |                               |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| Randomization/enrollment <sup>v</sup>                                       |                               | X                                               |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| IMP                                                                         |                               |                                                 |     |                                                                                                                       |        |         |     |         |     |         |                               |     |
| IV infusion of BNT324 <sup>w</sup>                                          |                               | X                                               |     |                                                                                                                       |        | X       |     | X       |     |         | X                             |     |
| IV infusion of BNT327 <sup>x</sup>                                          |                               | X                                               |     |                                                                                                                       |        | X       |     | X       |     |         | X                             |     |

|                                                                          | Screening period <sup>a</sup> | Treatment period                         |     |        |        |         |     |         |     |         |                               |     |
|--------------------------------------------------------------------------|-------------------------------|------------------------------------------|-----|--------|--------|---------|-----|---------|-----|---------|-------------------------------|-----|
|                                                                          |                               | Cycle 1 (21 days)                        |     |        |        | Cycle 2 |     | Cycle 3 |     |         | Cycle 4 and subsequent cycles |     |
|                                                                          |                               | D1                                       |     | D8     | D15    | D1      |     | D1      |     | D8      | D1                            |     |
|                                                                          |                               | BI                                       | EOI |        |        | BI      | EOI | BI      | EOI |         | BI                            | EOI |
| Visit window                                                             | D -28 to Day -1               | -                                        |     | ±1 day | ±1 day | ±3 days |     | ±3 days |     | ±3 days | ±3 or ±4 days <sup>b</sup>    |     |
| Additional evaluations                                                   |                               |                                          |     |        |        |         |     |         |     |         |                               |     |
| PK sampling                                                              |                               | See <a href="#">Table 14</a> for details |     |        |        |         |     |         |     |         |                               |     |
| ADA sampling                                                             |                               | See <a href="#">Table 14</a> for details |     |        |        |         |     |         |     |         |                               |     |
| Archival tumor tissue <sup>y</sup>                                       | X                             |                                          |     |        |        |         |     |         |     |         |                               |     |
| Fresh tumor biopsy (optional) <sup>y</sup>                               | (X)                           | As clinically indicated                  |     |        |        |         |     |         |     |         |                               |     |
| Blood biomarker samples <sup>z</sup>                                     |                               | X                                        |     |        |        |         |     | X       |     |         |                               |     |
| EORTC QLQ-C30, EORTC QLQ-LC29 (Part 2 Cohort 1 and 2 only) <sup>aa</sup> | X                             | X                                        |     |        |        | X       |     | X       |     |         | X                             |     |
| PGIS (Part 2 Cohort 1 and 2 only) <sup>aa</sup>                          |                               | X                                        |     |        |        | X       |     | X       |     |         | X                             |     |
| PGIC (Part 2 Cohort 1 and 2 only) <sup>aa</sup>                          |                               |                                          |     |        |        | X       |     | X       |     |         | X                             |     |

- a **Screening period:** Once informed consent is obtained, all screening procedures should be completed within 28 days before the first dose of IMP unless otherwise specified. For details, see Section 8.2 in CTP.
- b **Visit window in Cycle 4 and subsequent cycles:** The visit window is ±3 days for Cycle 4 to Cycle 8 and ±4 days for Cycle 9 and thereafter.
- c **Assessment of inclusion/exclusion criteria:** The investigator is advised to continuously assess the participant for potentially exclusionary conditions throughout the screening period. Compliance with all inclusion/exclusion criteria must be confirmed and documented immediately before randomization/enrollment.
- d **Tumor history/medical history:** This includes tumor history (including mutation status, PD-L1 expression status, TNM staging), medical history, concomitant and relevant past treatments such as prior radiotherapy and prior surgeries for primary disease, the allergy history, drug dependence history, history of alcohol use, and smoking history of participants. For details, see Section 8.2 in CTP.
- e **Ophthalmological examination:** For Cycle 3, assessments should be at ±10 days of C3D1. Assessment may be repeated if clinically indicated. For further details, see Section 8.4.1 in CTP.
- f **Physical examination:** For details, see Section 8.4.2 in CTP.
- g **ECOG PS scoring:** Will be performed before any blood draws or other invasive procedures during the screening period (within 7 days before the first dose of IMP), on Day 1 of each cycle starting from Cycle 1 (within 3 days BI), and when clinically indicated. Scoring items are provided in Section 8.4.4 in CTP.

- h **Body weight:** Will be measured during the screening period (within 7 days before the first dose), on Day 1 of each cycle (within 3 days BI), and when clinically indicated. The starting dose will be calculated using the body weight of the participant before the first dose. If the body weight of a participant varies by more than 10% of the baseline weight, the dose for the participant must be recalculated according to the updated body weight.
- i **Vital signs:** Vital signs will be evaluated within 7 days before the first dose during the screening period. If clinically indicated, additional assessments may be performed. On C1D1, C2D1, and C3D1, vital signs should be collected within 60 min BI of BNT324, 60 min after BNT324 EOI (BI of BNT327 infusion), and within 60 min after BNT327 EOI. On C1D8, C1D15, and C3D8, EOT visit and Safety Follow-up visit, the vital signs should be collected before any blood draws or other invasive procedures. Starting from Cycle 4, only pre-dose evaluation within 60 min BI of each investigational drug is required. For details, see Section 8.4.3 in CTP.
- j **SpO<sub>2</sub>:** Will be measured within 7 days before the first dose during the screening period, on Day 1 of each cycle within 1 day BI, and when clinically indicated. Where possible, SpO<sub>2</sub> will be measured before any blood draws or other invasive procedures. For details, see Section 8.4.9 in CTP.
- k **12-lead ECG:** Will be performed during the screening period (within 7 days before the first dose), prior to dosing on C1D1 and D1 of each subsequent cycle, at EOT visit and Safety Follow-up visit, and when clinically indicated. On C1D1, C2D1, and C3D1, ECG tests should be performed before infusion (within 90 min BI), and within 30 min after EOI of each investigational drug, and on C1D1 and C3D1 at 6 hours ( $\pm 1$  h) after the start of the infusion of BNT324. ECG tests will be performed after the participants have rested in supine/semi-supine position for at least 5 min. Where possible, ECGs should be done before any blood draws or other invasive procedures (if ECG is performed after blood collection, at least 1 h must elapse). At screening and in case of QTcF prolongation ( $>500$  ms) or any clinically significant abnormalities, the ECG should be performed in triplicate at least 1 min apart and within 5 min in total (unless contraindicated by the provision of emergency care) and re-evaluation should be confirmed by qualified personnel of the institution. For details, see Section 8.4.5 in CTP.
- l **Echocardiography:** Will be performed during the screening period, at C4D1 BI and then every 4 cycles during treatment, and as clinically indicated, up to 10 days BI. For details, see Section 8.4.6 in CTP.
- m **AE/SAE monitoring:** AEs and SAEs should be reported and graded according to NCI CTCAE v5.0 at each visit. AE and SAE collection will commence from the time when the participant signs the ICF and continue until 90 days after last dose of the IMP. Medical occurrences that begin before the start of trial treatment administration but after obtaining informed consent, will be collected as pre-existing conditions on the respective medical history/current medical conditions section of the CRF and not in the AE section. If the AE/SAE persist, they should be followed up as defined in Section 8.5.3 in CTP. Additionally, participants should be advised to immediately report cough, dyspnea, fever and/or any new or worsening respiratory or ocular symptoms. For details about AE/SAE reporting, see Section 8.5.2 and Section 8.5.4 in CTP.
- n **Prior and concomitant medications:** All prior and concomitant medications from the time when the ICF is signed to the Safety Follow-up visit should be recorded, including supportive therapeutic drugs (such as antiemetic treatment and prophylactic treatment as described in Section 6.8 in CTP), drugs for AE or disease treatment, and non-drug supportive interventions (such as blood transfusion). For guidance on prior and concomitant medications or procedures, see Section 8.4.10 in CTP.
- o **Pregnancy counseling:** Participants must adhere to the contraception requirements described in the protocol. For the details, see Section 8.5.5 and Section 10.5 in CTP.
- p **Hematology and blood biochemistry:** Will be performed within 7 days before the first dose, on C1D1, C1D8, C1D15, within 3 days BI at C2D1 and D1 of each subsequent cycle, and when clinically indicated. Sample collection should be performed before IMP administration. For details of the assessments including all assessed parameters, see Section 8.4.7 in CTP.
- q **Coagulation test, thyroid function test, myocardial enzyme test:** Will be performed within 7 days before the first dose, on C1D1, C1D8, C1D15, within 3 days BI at C2D1 and D1 of each subsequent cycle, and when clinically indicated. Sample collection should be performed before IMP administration. For details of the assessments including all assessed parameters, see Section 8.4.7 in CTP.
- r **Urinalysis:** Will be performed within 7 days before the first dose of IMP, on C1D8, within 3 days BI in C2D1 and D1 of each subsequent cycle, and when clinically indicated. Urinalysis should be performed before IMP administration. Participants with a 2+ or greater proteinuria by dipstick reading should undergo further assessment with a 24 h urine collection. For further details, see Section 8.4.7 in CTP.
- s **Pregnancy test:** For POCBP only. A serum  $\beta$ -hCG pregnancy test should be conducted within 7 days before the first dose of IMP. Serum or highly sensitive urine  $\beta$ -hCG pregnancy tests should also be performed within 3 days BI on Day 1 of every cycle, and if indicated. If a urine pregnancy test is positive, trial treatment should be withheld until a confirmatory serum pregnancy test is negative. If the confirmatory serum pregnancy test is positive, then trial treatment must be discontinued. Participants should be actively counseled regarding the required contraception. For further details about pregnancy testing and pregnancy counseling, see Section 8.5.5 and Section 10.5 in CTP.

- t **Serum virology:** Serum virology tests are valid for 180 days prior to enrollment if, at the investigator's discretion, no risk factors for HBV, HCV and HIV are present; otherwise, testing should be performed as planned. For the virology tests, see Section 8.4.7 in CTP.
- u **Tumor evaluation:** Participants will undergo tumor assessments at screening, every 6 weeks ( $\pm 7$  days) for the first 36 weeks after first dose of IMP, and every 12 weeks ( $\pm 14$  days) thereafter, regardless of dose delays, until disease progression, withdrawal of consent, or the start of next-line anticancer therapy (whichever occurs earlier). Tumor assessments should continue according to schedule in participants who discontinue treatment for other reasons. The same radiographic procedure used to assess disease sites at screening should be used for subsequent tumor assessments (e.g., the same protocol for a CT scan).  
For Part 1, participants with a PR or CR should undergo a confirmatory tumor assessment 4 to 7 weeks later, which can replace a scheduled assessment in the same time window. Thereafter, the investigator can schedule imaging according to local standard practice. If there is clinical suspicion of disease progression, an unscheduled scan should be performed before resuming scans at the original testing interval.  
For Part 2, participants with a PR or CR must undergo a confirmatory tumor assessment 4 to 7 weeks later, which can replace a scheduled assessment in the same time window. If the confirmatory scan shows a reduction in tumor burden, treatment and subsequent tumor assessments can continue as scheduled. If the confirmatory scan shows a stabilization of tumor burden (after initial progression), treatment continuation is allowed if the participant is deriving clinical benefit, after discussion with the medical monitor. After the confirmatory scan, subsequent tumor assessments must follow local standard practice.  
If participants discontinue treatment due to clinical disease progression, tumor assessment should be performed at the end of trial. For details of tumor evaluation, see Section 8.3.1 in CTP.
- v **Randomization/enrollment:** Randomization, where applicable, or enrollment and allocation of the participant into a specific cohort should occur after all screening procedures have been completed. By completing this step, the investigator confirms that at the time of enrollment, the participant meets all trial eligibility criteria. This should occur as close as possible to first dose and no longer than 7 days prior to first dose.
- w **IV infusion of BNT324:** BNT324 will be administered Q3W as IV infusion. The first BNT324 IV infusion should last at least 90 min. If the first infusion is well tolerated, the second and any subsequent infusion may last 60 min or more. For further details, see Section 6.3 and Section 8.4.11 in CTP.
- x **IV infusion of BNT327:** BNT327 will be administered Q3W as IV infusion 90 min after end of BNT324 infusion. For further details, see Section 6.3 and Section 8.4.11 in CTP.
- y **Tumor tissue:** FFPE samples should be collected during screening (note: for participants with SCLC, archival tissue should be provided, if possible, but it is not mandatory for enrollment). For participants who do not have archival tumor tissue collected during the previous 2 years, a newly obtained biopsy will be necessary. If more than one tumor tissue collection has been performed, the most recently collected sample is preferred. If the required tumor samples are not available and a fresh tumor biopsy is considered inappropriate, the sponsor's medical monitor should be consulted for determination of eligibility. Extra tissue can be submitted for additional biomarker research assessments. For further details, see Section 8.10 in CTP.
- z **Blood biomarker:** Blood biomarker samples for ctDNA testing will be collected as planned. For further details, see Section 8.10 in CTP.
- aa **HRQoL questionnaires (EORTC QLQ-C30, EORTC QLQ-LC29, PGIS, PGIC):** Questionnaires should be performed before any blood draws or other invasive procedures. At the end of trial treatment visit, the HRQoL questionnaires should be collected prior to informing participants that they are being withdrawn from the trial treatment to avoid introduction of bias in their response. For further details, see Section 8.6 in CTP.

Notes: As long as the absolute volume of blood drawn does not change per participant overall, the volume of individual blood draws may be adapted if required, e.g., due to unforeseen shortage of specific tubes. Leftover or back-up clinical sample aliquots of a participant, which have the exact same collection times, intended to a particular assessment can be used for another assay than that stated on the label if both assessments are described in the SoA and the participant has consented to all. This is only possible if a primary sample has been used first for its original purpose and it is confirmed that no re-testing is required. Samples can only be used interchangeably if the sample characteristics, collection, shipment, and storage conditions are the same.

Abbreviations: ADA = anti-drug antibody; AE = adverse event; BI = before infusion; CR = complete response; CT = computed tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group performance status; EOI = end of infusion; EOT = end of treatment; HBV = hepatitis B virus;  $\beta$ -hCG = beta-human chorionic gonadotropin;

FFPE = formalin-fixed paraffin-embedded; HCV = hepatitis C virus; HIV = human immunodeficiency virus; ICF = informed consent form; IMP = investigational medicinal product; IV = intravenous; MUGA = Multi-gated acquisition; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events of the National Cancer Institute; PD-L1 = programmed death-ligand 1; PGIC = Patient Global Impression of Change; PGIS = Patient Global Impression of Severity; PK = pharmacokinetics; POCBP = participants of childbearing potential; PR = partial response; QTcF = QT interval corrected according to Fridericia's formula; Q3W = every three weeks; SAE = serious adverse event; SCLC = small cell lung cancer; SpO<sub>2</sub> = peripheral oxygen saturation; TNM = tumor node metastasis (staging system).

**Table 13: End of treatment and follow-up**

|                                                                                                                                          | After treatment                                                                                                       |                               |                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------|
|                                                                                                                                          | EOT <sup>a</sup>                                                                                                      | Safety follow-up <sup>b</sup> | Survival follow-up <sup>c</sup><br>once every 3 months |
| Visit window                                                                                                                             | +7 days                                                                                                               | ±10 days                      | ±14 days                                               |
| General conditions and safety                                                                                                            |                                                                                                                       |                               |                                                        |
| Full physical examination <sup>d</sup>                                                                                                   | X                                                                                                                     | X                             |                                                        |
| Ophthalmological examination <sup>e</sup>                                                                                                | X                                                                                                                     |                               |                                                        |
| ECOG PS scoring <sup>f</sup>                                                                                                             | X                                                                                                                     | X                             |                                                        |
| Body weight <sup>g</sup>                                                                                                                 | X                                                                                                                     | X                             |                                                        |
| Vital signs <sup>h</sup>                                                                                                                 | X                                                                                                                     | X                             |                                                        |
| SpO <sub>2</sub> <sup>i</sup>                                                                                                            | X                                                                                                                     | X                             |                                                        |
| Single 12-lead ECG <sup>j</sup>                                                                                                          | X                                                                                                                     | X                             |                                                        |
| Triplicate 12-lead ECG                                                                                                                   | See Section 8.4.5 in CTP for details                                                                                  |                               |                                                        |
| Echocardiography/MUGA <sup>k</sup>                                                                                                       | X                                                                                                                     |                               |                                                        |
| AE / SAE monitoring <sup>l</sup>                                                                                                         | X                                                                                                                     |                               |                                                        |
| Prior and concomitant medications <sup>m</sup>                                                                                           | X                                                                                                                     |                               |                                                        |
| Survival status and subsequent therapies (new anticancer therapy, subsequent radiotherapy and/or surgical procedure for primary disease) | X                                                                                                                     | X                             | X                                                      |
| Laboratory tests                                                                                                                         |                                                                                                                       |                               |                                                        |
| Hematology and blood biochemistry <sup>n</sup>                                                                                           | X                                                                                                                     | X                             |                                                        |
| Coagulation, thyroid function test, and myocardial enzyme test <sup>o</sup>                                                              | X                                                                                                                     | X                             |                                                        |
| Urinalysis <sup>p</sup>                                                                                                                  | X                                                                                                                     | X                             |                                                        |
| Stool routine and occult blood test                                                                                                      | Conduct if deemed necessary by the investigator                                                                       |                               |                                                        |
| Pregnancy test <sup>q</sup>                                                                                                              | X                                                                                                                     | X                             |                                                        |
| Efficacy evaluation                                                                                                                      |                                                                                                                       |                               |                                                        |
| Tumor evaluation <sup>r</sup>                                                                                                            | Once every 6 weeks (±7 days) for the first 36 weeks after first dose of IMP, and every 12 weeks (±14 days) thereafter |                               |                                                        |
| Additional evaluations                                                                                                                   |                                                                                                                       |                               |                                                        |
| PK sampling                                                                                                                              | See <a href="#">Table 14</a> for details                                                                              |                               |                                                        |
| ADA sampling                                                                                                                             | See <a href="#">Table 14</a> for details                                                                              |                               |                                                        |
| Blood biomarker samples <sup>s</sup>                                                                                                     | X                                                                                                                     |                               |                                                        |
| Tumor biopsy (optional)                                                                                                                  | X                                                                                                                     |                               |                                                        |
| EORTC QLQ-C30, EORTC QLQ-LC29 (Part 2 Cohort 1 and 2 only) <sup>t</sup>                                                                  | X                                                                                                                     |                               |                                                        |
| PGIS (Part 2 Cohort 1 and 2 only) <sup>t</sup>                                                                                           | X                                                                                                                     |                               |                                                        |
| PGIC (Part 2 Cohort 1 and 2 only) <sup>t</sup>                                                                                           | X                                                                                                                     |                               |                                                        |

a **EOT:** Within +7 days of the decision to discontinue trial treatment.

b **Safety Follow-up:** At 90 days (±10 days) after the last dose of the IMP or up to 10 days before the start of a new anti-tumor therapy, whichever occurs first. Participants will return to the trial site for clinical and laboratory evaluations, review of prior and concomitant medications, and evaluation of treatment-related toxicity. Participants who show continued related toxicity at this timepoint will be followed up at least once every 4 weeks until the end of

- follow-up as defined in Section 10.4.3 in CTP. An EOT evaluation that has been carried out during this period can be regarded as a safety follow-up evaluation and no repeated evaluation is required.
- c **Survival Follow-up:** Follow-up on-site or by telephone will be performed once every 3 months ( $\pm 14$  days) from the Safety Follow-up visit until death, withdrawal of consent, lost to follow-up, or trial closure, whichever occurs first.
  - d **Physical examination:** Assessments may be repeated if clinically indicated. For details, see Section 8.4.2 in CTP.
  - e **Ophthalmological examination:** Assessments may be repeated if clinically indicated. For further details, see Section 8.4.1 in CTP.
  - f **ECOG PS scoring:** Will be performed at EOT visit, Safety Follow-up visit, and when clinically indicated. Scoring items are provided in Section 8.4.4 in CTP.
  - g **Body weight:** Will be measured as planned and when clinically indicated.
  - h **Vital signs:** Vital signs will be evaluated as planned and when clinically indicated. Where possible, measurements should be done before any blood draws or other invasive procedures. For details, see Section 8.4.3 in CTP.
  - i **SpO<sub>2</sub>:** SpO<sub>2</sub> will be evaluated as planned and when clinically indicated. Where possible, measurements should be done before any blood draws or other invasive procedures. For details, see Section 8.4.9 in CTP.
  - j **Single 12-lead ECG:** Single ECGs will be performed as planned and when clinically indicated. Single ECG are not collected if triplicate ECGs are required at the same timepoints per Section 8.4.5 in CTP. Where possible, ECGs should be done before any blood draws or other invasive procedures (if ECG is performed after blood collection, at least 1 h must elapse). In case of QTcF prolongation ( $>500$  ms), the ECG should be repeated and the average value recorded. For details, see Section 8.4.5 in CTP.
  - k **Echocardiography:** Will be performed at EOT and as clinically indicated. If echocardiography/MUGA are performed within 60 days of EOT, they do not need to be repeated, unless medical indicated (e.g., in case of AEs that could be due to new onset of cardiac dysfunction). For details, see Section 8.4.6 in CTP.
  - l **AE/SAE monitoring:** AEs and SAEs should be reported and graded according to NCI CTCAE v5.0 at each visit. AE and SAE collection will be done 90 days after last dose of the IMP. Medical occurrences that begin before the start of trial treatment administration but after obtaining informed consent, will be collected as pre-existing conditions on the respective medical history/current medical conditions section of the CRF and not in the AE section. Afterwards only related AE/SAE will be reported. If the AE/SAE persist, they should be followed up as defined in Section 8.5.3 in CTP. Additionally, participants should be advised to immediately report cough, dyspnea, fever and/or any new or worsening respiratory or ocular symptoms. For details about AE/SAE reporting, see Section 8.5.2 and Section 8.5.4 in CTP.
  - m **Prior and concomitant medications:** All prior and concomitant medications from the time when the ICF is signed to the Safety Follow-up visit should be recorded, including supportive therapeutic drugs (such as antiemetic treatment and prophylactic treatment as described in Section 6.8 in CTP), drugs for AE or disease treatment, and non-drug supportive interventions (such as blood transfusion). For guidance on prior and concomitant medications or procedures, see Section 8.4.10 in CTP.
  - n **Hematology and blood biochemistry:** Will be assessed as planned and when clinically indicated. For details of the assessments including all assessed parameters, see Section 8.4.7 in CTP.
  - o **Coagulation test, thyroid function test, myocardial enzyme test:** Will be assessed as planned and when clinically indicated. For details of the assessments including all assessed parameters, see Section 8.4.7 in CTP.
  - p **Urinalysis:** Will be assessed as planned and when clinically indicated. Participants with a 2+ or greater proteinuria by dipstick reading should undergo further assessment with a 24 h urine collection. For details of the assessments including all assessed parameters, see Section 8.4.7 in CTP.
  - q **Pregnancy test:** For POCBP, a serum or highly sensitive urine  $\beta$ -hCG pregnancy test should be performed as planned and if indicated. For further details about pregnancy testing and pregnancy counseling, see Section 8.5.5 and Section 10.5 in CTP.
  - r **Tumor evaluation:** Participants will undergo tumor assessments until disease progression, withdrawal of consent or the start of next-line anticancer therapy (whichever occurs earlier). Tumor assessments should continue according to schedule in participants who discontinue treatment for other reasons. Participants with a PR or CR should undergo a confirmatory tumor assessment 4 to 7 weeks later, which can replace a scheduled assessment in the same time window. The same radiographic procedure used to assess disease sites at screening should be used for subsequent tumor assessments (e.g., the same protocol for a CT scan). For details of tumor evaluation, see Section 8.3.1 in CTP.
  - s **Blood biomarker:** Blood biomarker samples for ctDNA testing will be collected as planned. For further details, see Section 8.10 in CTP.
  - t **HRQoL questionnaires (EORTC QLQ-C30, EORTC QLQ-LC29, PGIS, PGIC):** At the EOT visit, the HRQoL questionnaires should be collected prior to informing participants that they are being withdrawn from the trial treatment to avoid introduction of bias in their response.

## Notes:

As long as the absolute volume of blood drawn does not change per participant overall, the volume of individual blood draws may be adapted if required, e.g., due to unforeseen shortage of specific tubes. Leftover or back-up clinical sample



aliquots of a participant, which have the exact same collection times, intended to a particular assessment can be used for another assay than that stated on the label if both assessments are described in the SoA and the participant has consented to all. This is only possible if a primary sample has been used first for its original purpose and it is confirmed that no re-testing is required. Samples can only be used interchangeably if the sample characteristics, collection, shipment, and storage conditions are the same.

Abbreviations: ADA = anti-drug antibody; AE = adverse event; CR = complete response; CT = computed tomography; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group performance status; EOT = end of treatment;  $\beta$ -hCG = beta-human chorionic gonadotropin; ICF = informed consent form; IMP = investigational medicinal product; MUGA = Multi-gated acquisition; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events of the National Cancer Institute; PK = pharmacokinetics; POCBP = participants of childbearing potential; PR = partial response; QTcF = QT interval corrected according to Fridericia's formula; SAE = serious adverse event; SpO<sub>2</sub> = peripheral oxygen saturation.

**Table 14: Schedule of pharmacokinetic and anti-drug antibody**

| Sampling timepoint <sup>a</sup>                             |                         |                  | PK sample<br>BNT324 <sup>a, b</sup> | PK sample<br>BNT327 <sup>a, b</sup> | ADA sample<br>BNT324 <sup>a, b</sup> | ADA sample<br>BNT327 <sup>a, b</sup> |
|-------------------------------------------------------------|-------------------------|------------------|-------------------------------------|-------------------------------------|--------------------------------------|--------------------------------------|
| C1D1                                                        | Before infusion         | Within 8 h       | X                                   | X                                   | X                                    | X                                    |
|                                                             | After end of infusion   | Within 30 min    | X                                   | X                                   |                                      |                                      |
|                                                             | After start of infusion | 6 h $\pm$ 1 h    | X                                   |                                     |                                      |                                      |
| C1D8                                                        |                         | 168 h $\pm$ 24 h | X                                   |                                     |                                      |                                      |
| C1D15                                                       |                         | 336 h $\pm$ 24 h | X                                   |                                     | X                                    | X                                    |
| C2D1 $\pm$ 3 d                                              | Before infusion         | Within 8 h       | X                                   | X                                   | X                                    | X                                    |
|                                                             | After end of infusion   | Within 30 min    | X                                   | X                                   |                                      |                                      |
| C3D1 $\pm$ 3 d                                              | Before infusion         | Within 8 h       | X                                   |                                     |                                      |                                      |
|                                                             | After end of infusion   | Within 30 min    | X                                   |                                     |                                      |                                      |
|                                                             | After start of infusion | 6 h $\pm$ 1 h    | X                                   |                                     |                                      |                                      |
| C3D8                                                        |                         | 168 h $\pm$ 24 h | X                                   |                                     |                                      |                                      |
| C3D15 <sup>d</sup>                                          |                         | 336 h $\pm$ 24 h | X                                   |                                     |                                      |                                      |
| C4D1, C6D1,<br>C8D1 $\pm$ 3 d                               | Before infusion         | Within 8 h       | X                                   | X                                   | X                                    | X                                    |
|                                                             | After end of infusion   | Within 30 min    | X                                   | X                                   |                                      |                                      |
| C10D1, then<br>every 4 cycles<br>(C14D1, etc.)<br>until EOT | Before infusion         | Within 8 h       |                                     |                                     | X                                    | X                                    |
| EOT                                                         |                         |                  | X                                   |                                     | X                                    | X                                    |
| 90-d Safety Follow-up (90 d $\pm$ 10 d after last dosing)   |                         |                  | X                                   |                                     | X                                    |                                      |
| Survival Follow-up                                          |                         |                  |                                     |                                     | X <sup>c</sup>                       |                                      |
| Within 72 hours of occurrence of a DLT or SAE (if possible) |                         |                  | X                                   |                                     | X                                    |                                      |

- Timepoints for PK and ADA are to be linked to the relevant IMP (i.e., BNT324 or BNT327).
- PK assay may be performed on an ADA sample if a PK sample is not collected at that visit, and it is deemed necessary. The PK sample collection at the timepoint mentioned above is mandatory unless the sponsor gives prior approval.
- For participants with a positive ADA at any follow-up visit (or EOT visit if follow-up visit was not performed), an additional serum ADA sample will be collected every 3 months ( $\pm$  1 month) up to 1 year from the last dose of trial intervention, or if the ADA becomes negative, or if the ADA titer becomes less than baseline (applicable when pre-existing ADA is observed), or if the participant starts another therapy for cancer, or withdraws consent from the trial, or dies, whichever occurs first.
- The C3D15 PK sample will only be collected during Part 1 Dose Escalation.

Abbreviations: ADA = anti-drug antibody; CnDn = Cycle n Day n; d = day; DLT = dose-limiting toxicity; EOT = end of treatment; IMP = investigation medicinal product; PK = pharmacokinetics; SAE = serious adverse event.

## Appendix 2 BOIN Design

### **BOIN design in Part 1 (Dose Escalation)**

Part I will use the BOIN design ([Liu and Yuan 2015](#), [Lin and Yin 2017](#)) to find the RP2D ([Figure 2](#)). The BOIN design is a novel Bayesian dose finding method that optimizes participant treatment ethics by minimizing the chance of exposing participants to sub-therapeutic and overly toxic doses. The BOIN design is implemented in a simple way similar to the traditional 3+3 design but is more flexible and possesses superior operating characteristics that are comparable to those of the more complex model-based designs ([Lin and Yin 2017](#)). This trial was designed and will be conducted using Shiny app BOIN for combination trials.

The target toxicity rate is 0.27 and the maximum sample size is 54. Participants will be enrolled and treated in cohorts of 3. DLTs are defined in CTP v3.1 Section 6.3.1.3, and only those DLTs that occur within the first cycle will be used for dose finding. Let  $(j,k)$  denote the combination of  $j$ th dose level of BNT324 and  $k$ th dose level of BNT327, where  $j = 0, 1, 2$  and  $k = 0, 1$ . the BOIN design uses the following rule, optimized to minimize the probability of incorrect dose assignment, to guide dose escalation/de-escalation:

- If the observed DLT rate at the current dose is  $\leq 0.213$ , escalate the dose to one of the two next higher dose levels (i.e.,  $(j+1,k)$  or  $(j,k+1)$ ), using the rule as described below.
- If the observed DLT rate at the current dose is  $> 0.322$ , de-escalate the dose to one of the two next lower dose levels (i.e.,  $(j-1,k)$  or  $(j,k-1)$ ), using the rule as described below. Exception is made when 1/3 DLT occurs, at which point the decision is to stay at the current dose, in order to align with the common practice of the 3+3 design and allow additional data accumulation.
- Otherwise, stay at the current dose.

For the purpose of overdose control, doses  $(j,k)$  and higher levels will be eliminated from further examination if  $\Pr(p_{jk} > 0.27 \mid \text{data}) > 0.95$  and at least 3 participants have been treated at dose level  $(j,k)$ , where  $p_{jk}$  is the true DLT rate of dose level  $(j,k)$ . This posterior probability is evaluated based on the beta-binomial model  $y_{jk} \mid p_{jk} \sim \text{binomial}(p_{jk})$  with  $p_{jk} \sim \text{uniform}(0,1)$ , where  $y_{jk}$  is the number of participants experienced DLT at dose level  $(j,k)$ . When the lowest dose is eliminated, the trial will be stopped for safety. The probability cut-off 0.95 is chosen to be consistent with the common practice that when the target DLT rate  $\leq 1/6$ , a dose with 2/3 participants experienced DLT is eliminated. The above dose escalation/de-escalation and elimination rule can be equivalently presented in [Table 15](#), which will be used to conduct the trial.

The steps to implement the BOIN design are described as follows:

1. Participants in the first cohort are treated at dose combination (0,0).
2. Suppose that the current dose is  $(j,k)$ . To assign a dose to the next cohort of participants, conduct dose escalation/de-escalation according to the rule displayed in [Table 15](#). When using [Table 15](#), please note the following:
  - a. When the rule indicates dose escalation, escalate the dose to  $(j+1,k)$  or  $(j,k+1)$ , whichever has the highest desirability score as provided in [Table 16](#). When  $(j+1,k)$  and  $(j,k+1)$  have the same desirability score, randomly choose one of them for dose escalation or, or choose one at the discretion of the sponsor. Of note, if the

decision at the current dose combination  $(j,k)$ . is to escalate to a higher dose level, and the available dose escalation set contains untried dose combinations, the sponsor may also simultaneously assign one cohort of participants to  $(j+1,k)$  and one cohort to  $(j,k+1)$ , which leads to parallel paths. If these two paths recommend the same dose combination for the next assignment, these paths would be merged into a single path. If these two paths result in conflicting escalation/stay/de-escalation decisions, the next dose level will be determined by IRC review.

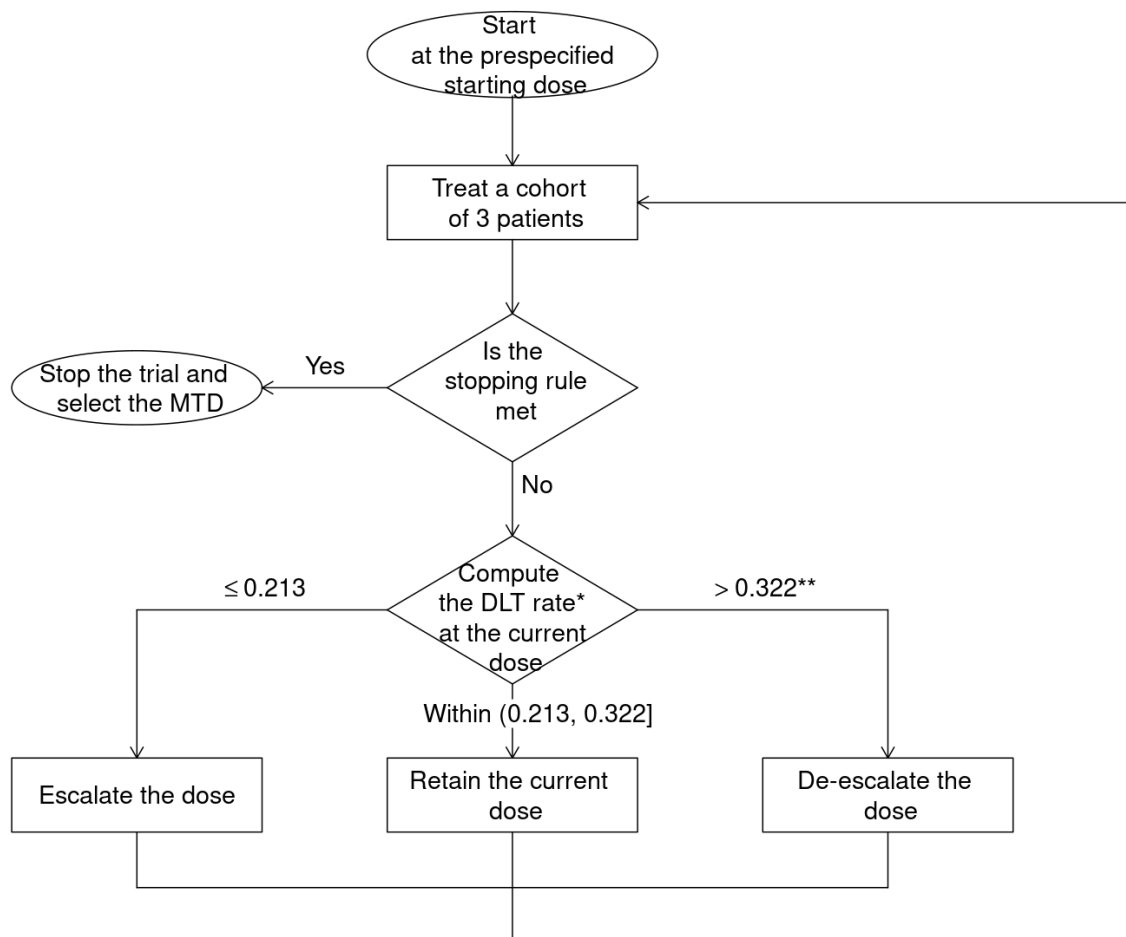
- b. When the rule indicates dose de-escalation, de-escalate the dose to  $(j-1,k)$  or  $(j,k-1)$ , whichever has the highest desirability score as provided in [Table 16](#). When  $(j-1,k)$  and  $(j,k-1)$  have the same desirability score, randomly choose one of them for dose de-escalation, or choose one at the discretion of the sponsor.
  - c. “Eliminate” means eliminate the current and higher doses, i.e., the dose set  $[(j^*, k^*); j^* \geq j \text{ and } k^* \geq k]$ , from the trial to prevent treating any future participants at these doses because they are overly toxic.
  - d. When eliminating a dose, automatically de-escalate the dose to one of the two next lower dose levels (i.e.,  $(j-1,k)$  or  $(j,k-1)$ ) as described in step b. When the lowest dose is eliminated, stop the trial for safety. In this case, no dose should be selected as the RP2D.
  - e. If none of the actions (i.e., escalation, de-escalation or elimination) is triggered, treat the new participants at the current dose combination.
  - f. If the current dose is the lowest dose and the rule indicates dose de-escalation, treat the new participants at the lowest dose unless the number of DLTs reaches the elimination boundary, at which point terminate the trial for safety.
  - g. If the current dose is the highest dose and the rule indicates dose escalation, treat the new participants at the highest dose.
  - h. Of note, [Table 15](#) provides a general guidance, and the sponsor may choose to stay at a specific dose level when the recommended decision is to escalate (e.g., where there are 1/5 or 1/6 DLT) based on totality of the data. In addition, the desirability score in [Table 16](#) will only serve as a guiding principle when choosing the next dose level, and the sponsor may override the choice based on the observed toxicity profile at the current dose.
3. Repeat Step 2 until the maximum sample size of 54 is reached or stop the trial if the number of participants treated at the current dose reaches nine and the decision according to Step 2 is to stay at the current dose. Per IRC decision, depending on cumulative safety data, alternative doses of BNT324 or BNT327 may be explored with up to nine DLT-evaluable participants for each dose level.

**Table 15: Dose escalation/de-escalation rule for the BOIN design**

| Decision                       | Number of evaluable participants treated at current dose |   |    |    |   |   |   |
|--------------------------------|----------------------------------------------------------|---|----|----|---|---|---|
|                                | 3                                                        | 4 | 5  | 6  | 7 | 8 | 9 |
| Escalate if # of DLT $\leq$    | 0                                                        | 0 | 1  | 1  | 1 | 1 | 1 |
| Stay if # of DLT =             | 1*                                                       | 1 | NA | NA | 2 | 2 | 2 |
| De-escalate if # of DLT $\geq$ | 2                                                        | 2 | 2  | 2  | 3 | 3 | 3 |
| Eliminate if # of DLT $\geq$   | 3                                                        | 3 | 3  | 4  | 4 | 5 | 5 |

Note. “# of DLT” is the number of participants with at least 1 DLT.

\* indicates the modification of the decision from “de-escalate” to “stay”, aligning with common practice when observing 1 DLT out of 3 participants.



$$* \text{ DLT rate} = \frac{\text{Total number of patients who experienced DLT at the current dose}}{\text{Total number of evaluable patients treated at the current dose}}$$

\*\* Except 1/3 DLT, under which retain the current dose

**Figure 2: Flowchart for trial conduct using the BOIN design**

### ***Desirability score table for trial conduct***

Table 16 serves as a guide for dose selection when making decisions of dose escalation and de-escalation. When the observed DLT data at the current combination dose level  $(j,k)$  indicate the need for dose escalation as per the BOIN rule in Table 15, the next two higher combination dose levels will be selected, namely  $(j+1,k)$  and  $(j,k+1)$ , provided they both exist and have not been eliminated from the trial. To make the choice for escalation, the next higher combination dose with the higher desirability score will be prioritized as indicated in Table 16. If these two next higher combination doses have

identical desirability scores, the selection may be made randomly or at the discretion of the investigator. Similarly, when the observed DLT data at the current combination dose level  $(j,k)$  indicate the need for dose de-escalation as per the BOIN rule in Table 15, the next two lower dose combination levels,  $(j-1,k)$  and  $(j,k-1)$ , will be considered and the one with the higher desirability score will be chosen. However, the sponsor may override the choice based on the observed toxicity profile at the current dose.

Given a specific DLT outcome data (e.g., one participant with DLT out of three participants), the desirability score is defined as the rank of that outcome among all possible DLT outcomes in terms of the posterior probability that the true DLT rate is located within the “stay” interval  $(0.213, 0.322]$ , i.e.,  $\Pr(0.213 < p_{jk} \leq 0.322 \mid y_{jk}, n_{jk})$ , where  $y_{jk}$  is the number of participants with DLT, and  $n_{jk}$  is the number of evaluable participants at dose combination  $(j,k)$ . Specifically, the process for calculating the desirability score is as follows: Given the maximum sample size  $N = 54$  (= number of cohorts \* cohort size), we enumerate all possible values of  $(y_{jk}, n_{jk})$ , namely  $(0,0)$ ,  $(0,1)$ ,  $(1,1)$ ,  $(0,2)$ ,  $(1,2)$ ,  $(2,2)$ , ...,  $(N,N)$ . For each of these values, we calculate the corresponding  $\Pr(0.213 < p_{jk} \leq 0.322 \mid y_{jk}, n_{jk})$ , resulting in a total of  $(N+1)(N+2)/2 = 1540$  values. A DLT outcome resulting in dose elimination according to Table 16 is assigned a value of negative infinity. These values are then sorted in ascending order, and the desirability score associated with a specific value of  $(y_{jk}, n_{jk})$  is defined as the rank of  $\Pr(0.213 < p_{jk} \leq 0.322 \mid y_{jk}, n_{jk})$  among the  $(N+1)(N+2)/2 = 1540$  sorted values. Thus, choosing a combination dose with the highest desirability score is equivalent to choosing a combination dose with the highest posterior probability of the true DLT rate being within the “stay” interval  $(0.213, 0.322]$ , i.e., the combination dose is more likely to be the RP2D.

**Table 16: Desirability score table ( $\phi = 0.27$ )**

| Num. Participants | Num. DLTs | Desirability Score | Num. Participants | Num. DLTs | Desirability Score | Num. Participants | Num. DLTs | Desirability Score |
|-------------------|-----------|--------------------|-------------------|-----------|--------------------|-------------------|-----------|--------------------|
| 0                 | 0         | 204                | 5                 | 0         | 196                | 8                 | 0         | 162                |
| 1                 | 0         | 227                | 5                 | 1         | 308                | 8                 | 1         | 275                |
| 1                 | 1         | 170                | 5                 | 2         | 279                | 8                 | 2         | 364                |
| 2                 | 0         | 224                | 5                 | $\geq 3$  | E                  | 8                 | 3         | 314                |
| 2                 | 1         | 233                | 6                 | 0         | 183                | 8                 | 4         | 221                |
| 2                 | 2         | 131                | 6                 | 1         | 304                | 8                 | $\geq 5$  | E                  |
| 3                 | 0         | 215                | 6                 | 2         | 319                | 9                 | 0         | 154                |
| 3                 | 1         | 272                | 6                 | 3         | 230                | 9                 | 1         | 261                |
| 3                 | 2         | 191                | 6                 | $\geq 4$  | E                  | 9                 | 2         | 367                |
| 3                 | $\geq 3$  | E                  | 7                 | 0         | 172                | 9                 | 3         | 352                |
| 4                 | 0         | 205                | 7                 | 1         | 291                | 9                 | 4         | 254                |
| 4                 | 1         | 298                | 7                 | 2         | 346                | 9                 | $\geq 5$  | E                  |
| 4                 | 2         | 237                | 7                 | 3         | 268                | NA                | NA        | NA                 |
| 4                 | $\geq 3$  | E                  | 7                 | $\geq 4$  | E                  | NA                | NA        | NA                 |

Note. “Num. participants” is the number of evaluable participants, “Num. DLT” is the number of participants who experience DLT, and “E” indicates that the dose level is too toxic and should be eliminated from the trial. If both dose levels should be eliminated, then the trial should stay at the current dose level.

It is important to note that while the numerical value of the desirability score may change with different maximum sample sizes  $N$ , the relative ranking between any two outcomes remains consistent regardless of the specific value of  $N$ . In our application, we present the desirability table for a maximum of  $m = 9$  evaluable participants only, where  $m$  represents the maximum number of participants that could potentially be assigned to a combination dose.

### Appendix 3 Censoring Rules for PFS and DOR

#### PFS:

Event and censoring rules for primary PFS analysis are included in [Table 17](#) below.

**Table 17: Event and censoring rules for PFS**

| Case No. | Case Scenario                                                                                                                                                                                                                            | Event / Censored | Event or Censoring Date                                                                                                                                                                                                                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | PD not immediately preceded by $\geq 2$ consecutive missed tumor assessments, and before new systemic anti-cancer therapy for primary disease, or before new radiotherapy for primary disease except for palliative treatment, if any    | Event            | Date of assessment of PD                                                                                                                                                                                                                                                                                                                |
| 2        | Death not immediately preceded by $\geq 2$ consecutive missed tumor assessments, and before new systemic anti-cancer therapy for primary disease, or before new radiotherapy for primary disease except for palliative treatment, if any | Event            | Date of death                                                                                                                                                                                                                                                                                                                           |
| 3        | No baseline tumor assessment                                                                                                                                                                                                             | Censored         | Date of first dose                                                                                                                                                                                                                                                                                                                      |
| 4        | No post-baseline tumor assessment                                                                                                                                                                                                        | Censored         | Date of first dose                                                                                                                                                                                                                                                                                                                      |
| 5        | PD or death immediately after $\geq 2$ consecutive missed tumor assessments, or after new systemic anti-cancer therapy for primary disease, or before new radiotherapy for primary disease except for palliative treatment, if any       | Censored         | Date of last evaluable tumor assessment prior to the earlier date of $\geq 2$ missed tumor assessments and new systemic anti-cancer therapy, if any<br>Or date of first dose if no post-baseline evaluable tumor assessment prior to the earlier date of $\geq 2$ missed tumor assessments and new systemic anti-cancer therapy, if any |

| Case No. | Case Scenario                                                                                                                                                    | Event / Censored | Event or Censoring Date                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6        | No PD at start of new systemic anti-cancer therapy for primary disease, or at the start of new radiotherapy for primary disease, except for palliative treatment | Censored         | Date of last evaluable tumor assessment prior to new systemic anti-cancer therapy or new radiotherapy, except for palliative treatment<br>Or date of first dose if no post-baseline evaluable tumor assessment prior to new systemic anti-cancer therapy |
| 7        | No PD or death at database cutoff date                                                                                                                           | Censored         | Date of last evaluable tumor assessment prior to database cutoff date                                                                                                                                                                                    |

For situations where multiple scenarios apply, the event/censor status of a participant should follow the order of scenario listed in the table.

### Two missed visits rule

The derivation to determine if 2 or more consecutive tumor assessments have been missed immediately before PD or death is based on the date of the earlier of PD or death and visit schedule of tumor assessments as indicated in SoA, that is, every 6 weeks (+/-7 days) for the first 48 weeks, and every 9 weeks (+/-7 days) thereafter.

If the duration between the earlier of PD/death date and the last post-baseline evaluable non-PD tumor assessment meets the below conditions, it will be defined as “PD or death immediately after two consecutive missed assessments”.

**Table 18: The definition for PD or death immediately after two consecutive missed assessments**

| Last evaluable post-baseline non-PD tumor assessment date (X1) <sup>a</sup>                | The earlier of death/PD date (X2) |
|--------------------------------------------------------------------------------------------|-----------------------------------|
| Missing, then set X1=Date of first dose of trial treatment                                 | If X2-X1 > 84+7                   |
| ≤ Tumor assessment visit of Week 30 (Day <sup>b</sup> ≤ 30*7+1+7)                          | If X2-X1 > 84+14                  |
| > Tumor assessment visit of Week 30 and ≤ Week 36 (30*7+1+7 < Day <sup>b</sup> ≤ 36*7+1+7) | If X2-X1 > (42+7) + (84+7)        |
| > Tumor assessment visit of Week 36 (Day <sup>b</sup> > 36*7+1+7)                          | If X2-X1 > 168+14                 |

<sup>a</sup> Include all scheduled and unscheduled tumor assessments.

<sup>b</sup> Day = Assessment Date – the date of first dose of trial treatment + 1

### DOR:

Event and censoring rules for DOR are included in [Table 19](#) below.



**Table 19: Event and Censoring Rules for DOR**

| Case No. | Case Scenario                                                                                                                                                                                                                            | Event / Censored        | Event or Censoring Date                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | PD not immediately preceded by $\geq 2$ consecutive missed tumor assessments, and before new systemic anti-cancer therapy for primary disease, or before new radiotherapy for primary disease except for palliative treatment, if any    | End of response (Event) | Date of assessment of PD                                                                                                                            |
| 2        | Death not immediately preceded by $\geq 2$ consecutive missed tumor assessments, and before new systemic anti-cancer therapy for primary disease, or before new radiotherapy for primary disease except for palliative treatment, if any | End of response (Event) | Date of death                                                                                                                                       |
| 3        | PD or death immediately after $\geq 2$ consecutive missed tumor assessments, or after new systemic anti-cancer therapy for primary disease, or after new radiotherapy for primary disease except for palliative treatment, if any        | Censored                | Date of last evaluable tumor assessment prior to the earlier date of $\geq 2$ missed tumor assessments and new systemic anti-cancer therapy, if any |
| 4        | No PD at start of new systemic anti-cancer therapy for primary disease, or at start of new radiotherapy for primary disease except for palliative treatment                                                                              | Censored                | Date of last evaluable tumor assessment prior to new systemic anti-cancer therapy                                                                   |
| 5        | No PD or death at database cutoff date                                                                                                                                                                                                   | Censored                | Date of last evaluable tumor assessment prior to database cutoff date                                                                               |

For situations where multiple scenarios apply, the event/censor status of a participant should follow the order of scenario listed in the table.

For the two missed visits rules refer to [Table 18](#).

## Appendix 4 BOR Rule

Table 20: BOR rules

| First Time Point Response | Subsequent Time Point Response                                                                                                                                                                                                                                                                                 | Confirmed Response* (Best Response) |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| CR                        | CR at least 4 weeks after initial CR, including CR, CR with the second CR at least 4 weeks ( $\geq 28$ days) after the initial CR, or multiple consecutive CRs (for example CR, CR, CR) with the last CR at least 4 weeks after initial CR [Below (CR, PR), (PR, PR), (PR, CR) scenarios follow same approach] | CR                                  |
| CR                        | CR less than 4 weeks after initial CR                                                                                                                                                                                                                                                                          | SD or NE (4)                        |
| CR                        | PR                                                                                                                                                                                                                                                                                                             | Query site on data accuracy         |
| CR                        | SD                                                                                                                                                                                                                                                                                                             | Query site on data accuracy         |
| CR                        | PD                                                                                                                                                                                                                                                                                                             | SD or PD (1)                        |
| CR                        | NE or No further evaluation                                                                                                                                                                                                                                                                                    | SD or NE (2)                        |
| CR                        | One NE prior to CR (at least 4 weeks between two CRs) **                                                                                                                                                                                                                                                       | CR                                  |
| PR                        | CR at least 4 weeks after initial PR                                                                                                                                                                                                                                                                           | PR                                  |
| PR                        | CR less than 4 weeks after initial PR                                                                                                                                                                                                                                                                          | SD or NE (4)                        |
| PR                        | PR at least 4 weeks after initial PR                                                                                                                                                                                                                                                                           | PR                                  |
| PR                        | PR less than 4 weeks after initial PR                                                                                                                                                                                                                                                                          | SD or NE (4)                        |
| PR                        | SD (3)                                                                                                                                                                                                                                                                                                         | SD or NE (4)                        |
| PR                        | PD                                                                                                                                                                                                                                                                                                             | SD or PD (1)                        |
| PR                        | NE or No further evaluation                                                                                                                                                                                                                                                                                    | SD or NE (2)                        |
| PR                        | One or two NE or one SD or one NE plus one SD prior to PR/CR (at least 4 weeks between two PRs, or PR and CR) **                                                                                                                                                                                               | PR                                  |
| SD                        | CR                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| SD                        | PR                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| SD                        | SD                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| SD                        | NE or No further evaluation                                                                                                                                                                                                                                                                                    | SD or NE (2)                        |
| SD                        | PD                                                                                                                                                                                                                                                                                                             | SD or PD (1)                        |
| PD                        | Any or No further evaluation                                                                                                                                                                                                                                                                                   | PD                                  |
| NE                        | CR                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| NE                        | PR                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| NE                        | SD                                                                                                                                                                                                                                                                                                             | SD or NE (4)                        |
| NE                        | NE                                                                                                                                                                                                                                                                                                             | NE                                  |
| NE                        | PD                                                                                                                                                                                                                                                                                                             | PD                                  |

| First Time Point Response | Subsequent Time Point Response     | Confirmed Response* (Best Response) |
|---------------------------|------------------------------------|-------------------------------------|
| NE                        | NED                                | NED                                 |
| NED                       | NED or PD or No further evaluation | NED                                 |

1) CR=complete response, PR=partial response, SD=stable disease, PD=progressive disease, NE=not evaluable, NED=no evidence of disease.

Non-CR/Non-PD is treated as equivalent as SD.

\* A best response of SD can only be made after the participant is on trial for a minimum of 6 weeks (-1 week window) (35 days), any tumor assessment indicating stable disease before this time period will have a Best Response of NE, unless PD is identified.

\*\* Subsequent documentation of CR may provide confirmation of a previously identified CR for participants where the second overall response was NE. Subsequent documentation of PR may provide confirmation of a previously identified PR for participants where the second (or more) overall response was NE. If the third time point response (TPR) confirms the CR (or PR) then the Confirmed Response will be CR (or PR). For this trial, only one intervening NE is allowed between CRs and up to two NE or one SD or one NE plus one SD is allowed between PRs (or PR followed by CR), e.g., CR-NE-CR = BOR of CR; PR-NE-PR = BOR of PR; PR-SD-PR = BOR of PR; PR-NE-NE-PR = BOR of PR. PR-NE-SD-PR = BOR of PR, PR-SD-NE-PR = BOR of PR.

**Note:** Best response will be CR or PR if the interval between the two CR/PR is  $\geq 4$  weeks (28 days).

Otherwise, the best response will be SD if the later CR/PR meets the 35-day requirement or NE otherwise.

(1) Best response will be SD if the first TPR is  $\geq 5$  weeks (6 weeks -1 week window) (35 days). Otherwise, the best response will be PD.

(2) Best response will be SD if the first TPR is  $\geq 5$  weeks (6 weeks -1 week window) (35 days). Otherwise, the best response will be NE.

(3) TPR is SD if the increase from the first to the second assessment does not qualify for PD.

(4) Best response will be SD if the second TPR is  $\geq 5$  weeks (6 weeks -1 week window) (35 days). Otherwise, the best response will be NE.

## Appendix 5 NCI-CTCAE V5.0 Toxicity Grading

Grade is classified based on the numeric lab result criteria in NCI CTCAE v5.0 grading (NCI-CTCAE V5.0 Toxicity/AE Grading criteria). The clinical criteria or condition will not be used.

**Table 21: NCI CTCAE V5 grading criteria for selected laboratory parameters for programming**

| Parameter                      | Direction | Standard Unit   | Grade 1                                                                                                  | Grade 2                                                                                               | Grade 3                                                                                                | Grade 4                                                                |
|--------------------------------|-----------|-----------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Hematology                     |           |                 |                                                                                                          |                                                                                                       |                                                                                                        |                                                                        |
| Hemoglobin                     | Decreased | g/L             | $100 \leq \text{Value} < \text{LLN}$                                                                     | $80 \leq \text{Value} < 100$                                                                          | $\text{Value} < 80$                                                                                    | -                                                                      |
|                                | Increased | g/L             | $\text{ULN} < \text{Value} \leq \text{ULN} + 20$                                                         | $\text{ULN} + 20 < \text{Value} \leq \text{ULN} + 40$                                                 | $\text{ULN} + 40 < \text{Value}$                                                                       | -                                                                      |
| Platelet count                 | Decreased | $10^9/\text{L}$ | $75 \leq \text{Value} < \text{LLN}$                                                                      | $50 \leq \text{Value} < 75$                                                                           | $25 \leq \text{Value} < 50$                                                                            | $\text{Value} < 25$                                                    |
| White blood cell count         | Decreased | $10^9/\text{L}$ | $3 \leq \text{Value} < \text{LLN}$                                                                       | $2 \leq \text{Value} < 3$                                                                             | $1 \leq \text{Value} < 2$                                                                              | $\text{Value} < 1$                                                     |
|                                | Increased | $10^9/\text{L}$ | -                                                                                                        | -                                                                                                     | $100 < \text{Value}$                                                                                   | -                                                                      |
| Absolute neutrophil count      | Decreased | $10^9/\text{L}$ | $1.5 \leq \text{Value} < \text{LLN}$                                                                     | $1 \leq \text{Value} < 1.5$                                                                           | $0.5 \leq \text{Value} < 1$                                                                            | $\text{Value} < 0.5$                                                   |
| Absolute lymphocyte count      | Decreased | $10^9/\text{L}$ | $0.8 \leq \text{Value} < \text{LLN}$                                                                     | $0.5 \leq \text{Value} < 0.8$                                                                         | $0.2 \leq \text{Value} < 0.5$                                                                          | $\text{Value} < 0.2$                                                   |
|                                | Increased | $10^9/\text{L}$ | -                                                                                                        | $4 < \text{Value} \leq 20$                                                                            | $20 < \text{Value}$                                                                                    | -                                                                      |
| Absolute eosinophils count     | Increased |                 | $\text{Value} > \text{ULN}$ and $> \text{baseline}$                                                      |                                                                                                       |                                                                                                        |                                                                        |
| Chemistry                      |           |                 |                                                                                                          |                                                                                                       |                                                                                                        |                                                                        |
| Blood glucose                  | Decreased | mmol/L          | $3 \leq \text{Value} < \text{LLN}$                                                                       | $2.2 \leq \text{Value} < 3$                                                                           | $1.7 \leq \text{Value} < 2.2$                                                                          | $\text{Value} < 1.7$                                                   |
| Alanine aminotransferase (ALT) | Increased | U/L             | $\text{ULN} < \text{Value} \leq 3 \times \text{ULN}$ , if baseline is normal                             | $3 \times \text{ULN} < \text{Value} \leq 5 \times \text{ULN}$ , if baseline is normal                 | $5 \times \text{ULN} < \text{Value} \leq 20 \times \text{ULN}$ , if baseline is normal                 | $20 \times \text{ULN} < \text{Value}$ , if baseline is normal          |
|                                |           |                 | $1.5 \times \text{baseline} \leq \text{Value} \leq 3.0 \times \text{baseline}$ , if baseline is abnormal | $3.0 \times \text{baseline} < \text{Value} \leq 5.0 \times \text{baseline}$ , if baseline is abnormal | $5.0 \times \text{baseline} < \text{Value} \leq 20.0 \times \text{baseline}$ , if baseline is abnormal | $20.0 \times \text{baseline} < \text{Value}$ , if baseline is abnormal |
| Albumin                        | Decreased | g/L             | $30 \leq \text{Value} < \text{LLN}$                                                                      | $20 \leq \text{Value} < 30$                                                                           | $\text{Value} < 20$                                                                                    | -                                                                      |

| Parameter                                   | Direction | Standard Unit             | Grade 1                                                                                  | Grade 2                                                                                                     | Grade 3                                                                              | Grade 4                                                 |
|---------------------------------------------|-----------|---------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| Alkaline phosphatase (ALP)                  | Increased | U/L                       | ULN < Value $\leq$ 2.5 $\times$ ULN, if baseline is normal                               | 2.5 $\times$ ULN < Value $\leq$ 5.0 $\times$ ULN, if baseline is normal                                     | 5.0 $\times$ ULN < Value $\leq$ 20.0 $\times$ ULN, if baseline is normal             | 20.0 $\times$ ULN < Value, if baseline is normal        |
|                                             |           |                           | 2.0 $\times$ baseline $\leq$ Value $\leq$ 2.5 $\times$ baseline, if baseline is abnormal | 2.5 $\times$ baseline < Value $\leq$ 5.0 $\times$ baseline, if baseline is abnormal                         | 5.0 $\times$ baseline < Value $\leq$ 20.0 $\times$ baseline, if baseline is abnormal | 20.0 $\times$ baseline < Value, if baseline is abnormal |
| Aspartate aminotransferase (AST)            | Increased | U/L                       | ULN < Value $\leq$ 3 $\times$ ULN, if baseline is normal                                 | 3 $\times$ ULN < Value $\leq$ 5 $\times$ ULN, if baseline is normal                                         | 5 $\times$ ULN < Value $\leq$ 20 $\times$ ULN, if baseline is normal                 | 20 $\times$ ULN < Value, if baseline is normal          |
|                                             |           |                           | 1.5 $\times$ baseline $\leq$ Value $\leq$ 3.0 $\times$ baseline, if baseline is abnormal | 3.0 $\times$ baseline < Value $\leq$ 5.0 $\times$ baseline, if baseline is abnormal                         | 5.0 $\times$ baseline < Value $\leq$ 20.0 $\times$ baseline, if baseline is abnormal | 20.0 $\times$ baseline < Value, if baseline is abnormal |
| Bicarbonate                                 | Decreased |                           | Value < LLN                                                                              | -                                                                                                           | -                                                                                    | -                                                       |
| Total bilirubin                             | Increased | umol/L                    | ULN < Value $\leq$ 1.5 $\times$ ULN, if baseline is normal                               | 1.5 $\times$ ULN < Value $\leq$ 3.0 $\times$ ULN, if baseline is normal                                     | 3.0 $\times$ ULN < Value $\leq$ 10.0 $\times$ ULN, if baseline is normal             | 10.0 $\times$ ULN < Value, if baseline is normal        |
|                                             |           |                           | 1.0 $\times$ baseline < Value $\leq$ 1.5 $\times$ baseline, if baseline is abnormal      | 1.5 $\times$ baseline < Value $\leq$ 3.0 $\times$ baseline, if baseline is abnormal                         | 3.0 $\times$ baseline < Value $\leq$ 10.0 $\times$ baseline, if baseline is abnormal | 10.0 $\times$ baseline < Value, if baseline is abnormal |
| Calcium (ionized)                           | Decreased | mmol/L                    | 1 $\leq$ Value < LLN                                                                     | 0.9 $\leq$ Value < 1.0                                                                                      | 0.8 $\leq$ Value < 0.9                                                               | Value < 0.8                                             |
|                                             | Increased | mmol/L                    | ULN < Value $\leq$ 1.5                                                                   | 1.5 < Value $\leq$ 1.6                                                                                      | 1.6 < Value $\leq$ 1.8                                                               | 1.8 < Value                                             |
| Corrected serum calcium                     | Decreased | mmol/L                    | 2.0 $\leq$ Value < LLN                                                                   | 1.75 $\leq$ Value < 2.0                                                                                     | 1.5 $\leq$ Value < 1.75                                                              | Value < 1.5                                             |
|                                             | Increased | mmol/L                    | ULN < Value $\leq$ 2.9                                                                   | 2.9 < Value $\leq$ 3.1                                                                                      | 3.1 < Value $\leq$ 3.4                                                               | 3.4 < Value                                             |
| Creatinine                                  | Increased | umol/L                    | ULN < Value $\leq$ 1.5 $\times$ ULN                                                      | 1.5 $\times$ ULN < Value $\leq$ 3 $\times$ ULN;<br>1.5 $\times$ baseline < Value $\leq$ 3 $\times$ baseline | 3 $\times$ ULN < Value $\leq$ 6 $\times$ ULN;<br>3 $\times$ baseline < Value         | 6 $\times$ ULN < Value                                  |
| estimated Glomerular Filtration Rate (eGFR) | Decreased | ml/min/1.73m <sup>2</sup> | 60 $\leq$ Value < LLN                                                                    | 30 $\leq$ Value $\leq$ 59                                                                                   | 15 $\leq$ Value $\leq$ 29                                                            | Value < 15                                              |

| Parameter                                        | Direction | Standard Unit | Grade 1                                                                             | Grade 2                                                                             | Grade 3                                                                              | Grade 4                                                 |
|--------------------------------------------------|-----------|---------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| Gamma glutamyl transferase                       | Increased |               | ULN < Value $\leq$ 2.5 $\times$ ULN, if baseline is normal                          | 2.5 $\times$ ULN < Value $\leq$ 5.0 $\times$ ULN, if baseline is normal             | 5.0 $\times$ ULN < Value $\leq$ 20.0 $\times$ ULN, if baseline is normal             | 20.0 $\times$ ULN < Value, if baseline is normal        |
|                                                  |           |               | 2.0 $\times$ baseline < Value $\leq$ 2.5 $\times$ baseline, if baseline is abnormal | 2.5 $\times$ baseline < Value $\leq$ 5.0 $\times$ baseline, if baseline is abnormal | 5.0 $\times$ baseline < Value $\leq$ 20.0 $\times$ baseline, if baseline is abnormal | 20.0 $\times$ baseline < Value, if baseline is abnormal |
| Magnesium                                        | Decreased | mmol/L        | 0.5 $\leq$ Value < LLN                                                              | 0.4 $\leq$ Value < 0.5                                                              | 0.3 $\leq$ Value < 0.4                                                               | Value < 0.3                                             |
|                                                  | Increased | mmol/L        | ULN < Value $\leq$ 1.23                                                             | -                                                                                   | 1.23 < Value $\leq$ 3.3                                                              | 3.3 < Value                                             |
| Potassium                                        | Decreased | mmol/L        | 3 $\leq$ Value < LLN                                                                | -                                                                                   | 2.5 $\leq$ Value < 3                                                                 | Value < 2.5                                             |
|                                                  | Increased | mmol/L        | ULN < Value $\leq$ 5.5                                                              | 5.5 < Value $\leq$ 6                                                                | 6 < Value $\leq$ 7                                                                   | 7 < Value                                               |
| Total protein                                    |           |               |                                                                                     |                                                                                     |                                                                                      |                                                         |
| Sodium                                           | Decreased | mmol/L        | 130 $\leq$ Value < LLN                                                              | 125 $\leq$ Value < 130                                                              | 120 $\leq$ Value < 125                                                               | Value < 120                                             |
|                                                  | Increased | mmol/L        | ULN < Value $\leq$ 150                                                              | 150 < Value $\leq$ 155                                                              | 155 < Value $\leq$ 160                                                               | 160 < Value                                             |
| Lactate dehydrogenase (LDH)                      | Increased | U/L           | ULN < Value                                                                         | -                                                                                   | -                                                                                    | -                                                       |
| Lipase                                           | Increased | mmol/L        | ULN < Value $\leq$ 1.5 $\times$ ULN                                                 | 1.5 $\times$ ULN < Value $\leq$ 2 $\times$ ULN                                      | 2 $\times$ ULN < Value $\leq$ 5 $\times$ ULN                                         | 5.0 $\times$ ULN < Value                                |
| Amylase                                          | Increased | mmol/L        | ULN < Value $\leq$ 2 $\times$ ULN                                                   | 2 $\times$ ULN < Value $\leq$ 5 $\times$ ULN                                        | 5 $\times$ ULN < Value                                                               | -                                                       |
| Creatine kinase                                  | Increased |               | >ULN - 2.5 $\times$ ULN                                                             | >2.5 $\times$ ULN - 5 $\times$ ULN                                                  | >5 $\times$ ULN - 10 $\times$ ULN                                                    | >10 $\times$ ULN                                        |
| Coagulation                                      |           |               |                                                                                     |                                                                                     |                                                                                      |                                                         |
| Prothrombin international normalized ratio (INR) | Increased | -             | 1.2 < Value $\leq$ 1.5                                                              | 1.5 < Value $\leq$ 2.5                                                              | 2.5 < Value                                                                          | -                                                       |
| Activated partial thromboplastin time (APTT)     | Increased | sec           | ULN < Value $\leq$ 1.5 $\times$ ULN                                                 | 1.5 $\times$ ULN < Value < 2.5 $\times$ ULN                                         | 2.5 $\times$ ULN < Value                                                             | -                                                       |

| Parameter  | Direction | Standard Unit | Grade 1                                                                                                       | Grade 2                                                                                                                      | Grade 3                                                                                                                      | Grade 4                                                                            |
|------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Fibrinogen | Decreased |               | $0.75 \times \text{LLN} \leq \text{Value} < 1.0 \times \text{LLN}$ ; if abnormal, <25% decrease from baseline | $0.5 \times \text{LLN} \leq \text{Value} < 0.75 \times \text{LLN}$ ; if abnormal, $\geq 25\%$ - < 50% decrease from baseline | $0.25 \times \text{LLN} \leq \text{Value} < 0.5 \times \text{LLN}$ ; if abnormal, $\geq 50\%$ - < 75% decrease from baseline | Value < $0.25 \times \text{LLN}$ ; if abnormal, $\geq 75\%$ decrease from baseline |

LLN = lower limit of normal; ULN = upper limit of normal.



## Appendix 6 Clinical Significance Criteria for Safety Parameters

Clinical significance of safety parameter results will be determined by the investigator using local reference ranges and documented in the eCRF.

## Appendix 7 Date Imputation Rules

**Table 22: Data imputation rules**

| Missing or Partial Date Variable | Imputation Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First dose of trial drug date    | No imputation will be performed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Last dose of trial drug date     | <p>If the last date of trial drug is completely missing and there is no End of Treatment (EoT) eCRF page completed and no death date, the participant should be ongoing and use the cut-off date for the analysis as the last dosing date. Note: Confirm that the participant is actively receiving dose at the time of the data cut-off.</p> <p>If the last date of trial drug is completely or partially missing and there is EITHER an EoT eCRF page completed OR a death date available (within the data cut-off date), then use this date as the last dose date:</p> <p>If only Year is available and Year &lt; Year of min (EoT date, death date), then use 31DECYYYY,</p> <p>If both year and month are available and Year = Year of min (EoT date, death date) and Month &lt; Month of min (EoT date, death date), then use = last day of the month.</p> <p>For all other conditions, use min (EoT date, death date).</p> |

| Missing or Partial Date Variable       | Imputation Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New anti-cancer therapy start date     | <p>First compute the temporary imputed date = max (PD date by investigator + 1 day, last dose date + 1 day):</p> <p>If the day is missing but month and year are available, the start date will be imputed as.</p> <p>if year &lt; the year of the temporary imputed date, then use (year, month, day = the last day of the month);</p> <p>if year &gt; the year of the temporary imputed date, then use date (year, month, day = 01)</p> <p>if year = the year of the temporary imputed date, then</p> <p>if (year, month) &lt; (year, month) of the temporary imputed date, then use last day of the month.</p> <p>if (year, month) = (year, month) of the temporary imputed date, then use the temporary imputed date</p> <p>if (year, month) &gt; (year, month) of the temporary imputed date, then use first day of the month</p> <p>If both the month and day are missing, the start date will be imputed as</p> <p>if year &lt; the year of the temporary imputed date, then use December 31<sup>st</sup> of the year</p> <p>if year = the year of the temporary imputed date, the use date of temporary imputed date</p> <p>if year &gt; the year of the temporary imputed date, then use January 1<sup>st</sup> of the year</p> <p>If the entire date is missing, the start date will be imputed as the temporary imputed date.</p> <p>If the resulting imputed value is after the end date of new anti-cancer therapy, then use the end date of the new anti-cancer therapy.</p> <p>Note: For participants who have not discontinued trial treatment at the analysis cut-off date, the last dose of trial treatment is set to the analysis cut-off date for the purpose of the above imputations.</p> |
| New anti-cancer therapy end date       | <p>If the day is missing but month and year are available, the end date will be imputed using the last day of the month.</p> <p>If both the month and day are missing, the end date will be imputed using the last day of the year (December 31).</p> <p>If the entire date is missing, no imputation is performed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Prior anti-cancer therapies start date | <p>If the day is missing but month and year are available, the medication/therapy start date day will be imputed as first day of the month.</p> <p>If day and month are missing, the medication/therapy start date will be imputed using January 1.</p> <p>If the start date year is missing, then the date is imputed as 1 day prior to the date of first dose of trial treatment.</p> <p>Note: If complete end date is available and the imputed start date is greater than the end date, then the start date should be set to the end date.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Missing or Partial Date Variable     | Imputation Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prior anti-cancer therapies end date | <p>If the day is missing but month and year are available, the medication/therapy end date day will be imputed using the last day of the month.</p> <p>If day and month are missing, the medication/therapy end date day and month will be imputed using the end of the year (31DECYYYY).</p> <p>If the end date year is missing, then impute with the date of first dose of trial treatment.</p> <p>Note: If the imputed end date is later than the date of first dose of trial treatment, then impute with the date of first dose of trial treatment.</p>                                                                                                                                                                                                                                                                                     |
| Last known alive date                | <p>Partial last known alive date will be only imputed if there is no documented death for a participant, the algorithm is:</p> <p>Missing day: Firstly, the first day of the month is imputed; Secondly, last known alive date = max (the maximal known visit/assessment date in the raw data, the temporarily imputed last known alive date);</p> <p>Missing month: Firstly, January 01 is imputed; Secondly, last known alive date = max (the maximal known visit/assessment date in the raw data, the temporarily imputed last known alive date);</p> <p>Missing year: last known alive date = the maximal known visit/assessment date in the raw data.</p>                                                                                                                                                                                  |
| Death date                           | <p>If the day is missing but month and year are available, the death date will be imputed to the maximum of (date of last known alive +1, the 1<sup>st</sup> day of the month and year of death).</p> <p>If day and month are missing, the death date will be imputed as the maximum (date of last known alive + 1, January 1<sup>st</sup> of the year of death).</p> <p>If day, month, and year are missing, the death date will be imputed as the day after the date of last known alive.</p>                                                                                                                                                                                                                                                                                                                                                 |
| Adverse event start date             | <p>If the day is missing but month and year are available, the AE start date will be imputed using the max (first day of the month, inform consent date), except when (year, month) = (year, month) of the first dose date, then the first dose date will be used instead.</p> <p>If day and month are missing, the AE start date will be imputed using max (January 01 of the year, inform consent date), except when year = year of the first dose date, then the first dose date will be used instead.</p> <p>If day, month, and year are missing, then it is imputed with the first dose date if the first dose date ≤ AE stop date or AE stop date is missing, else impute to the inform consent date.</p> <p>Note: If the imputed date is later than the AE stop date (possibly imputed), then the AE stop date will be used instead.</p> |
| Adverse event stop date              | <p>If the day is missing but month and year are available, the AE stop date will be imputed using the last day of the month.</p> <p>If day and month are missing, the AE stop date will be imputed using December 31.</p> <p>If day, month, and year are missing, the AE stop date will not be imputed.</p> <p>Note: If the imputed date is after the date of death, then the date of death will be used instead.</p>                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Missing or Partial Date Variable                           | Imputation Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Medication/therapy (excluding cancer therapies) start date | <p>If the day is missing but month and year are available, the medication/therapy start date day will be imputed as first day of the month.</p> <p>If day and month are missing, the medication/therapy start date will be imputed using January 1.</p> <p>If the start date year is missing, then the date is imputed as 1 day prior to trial treatment start date.</p> <p>Note: If complete end date is available and the imputed start date is greater than the end date, then the start date should be set to the end date.</p>                                                     |
| Medication/therapy (excluding cancer therapies) end date   | <p>If the day is missing but month and year are available, the medication/therapy end date day will be imputed using the last day of the month.</p> <p>If day and month are missing, the medication/therapy end date day and month will be imputed using the end of the year (31DECYYYY).</p> <p>If the end date year is missing, then the end date will not be imputed, but for classification into prior or concomitant, the medication will be considered as concomitant.</p> <p>Note: If the imputed end date is later than death date, then impute the end date to death date.</p> |

## Appendix 8 Mapping Rules For Countries Into Regions

Table 23: Mapping rules for region

| Region        | Country (based on recruiting site location) |
|---------------|---------------------------------------------|
| Asia-Pacific  | China, Taiwan, South Korea                  |
| North America | US                                          |
| Europe        | Turkey, France, Italy, Spain, Poland, UK    |
| Rest of World | Australia                                   |

# Signature page

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|------------------------|--------------------------------------|----------------------------|
| 2026-03-04<br>15:30:12 | Biostatistics<br>Approval            | shanjun.helian@biontech.us |
| 2026-03-04<br>19:34:9  | Clinical<br>Approval                 | holger.thurm@biontech.com  |
| 2026-03-05<br>4:5:38   | Clinical<br>Pharmacology<br>Approval | shuang.liang@biontech.us   |